

Lecture 05: Bayes and Bootstrap

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Textbook:

B Efron & T Hastie (2016) Computer Age Statistical Inference,
Cambridge University Press.

Abstract:

This lecture briefly reviews Bayesian reasoning and introduces the **Bayesian Information Criterion (BIC)** for generalized linear models.

Next it considers the problem of **false discovery rates**, which arises in modern large-scale analyses that test thousands of hypotheses concurrently. Fortunately, the issue has an elegant Bayesian solutions.

Finally, it introduces **permutational methods**, in particular **bootstrapping** for estimating standard error and confidence intervals, and **shuffling** to generate null distributions.

These important methods will be discussed further in the following lectures.

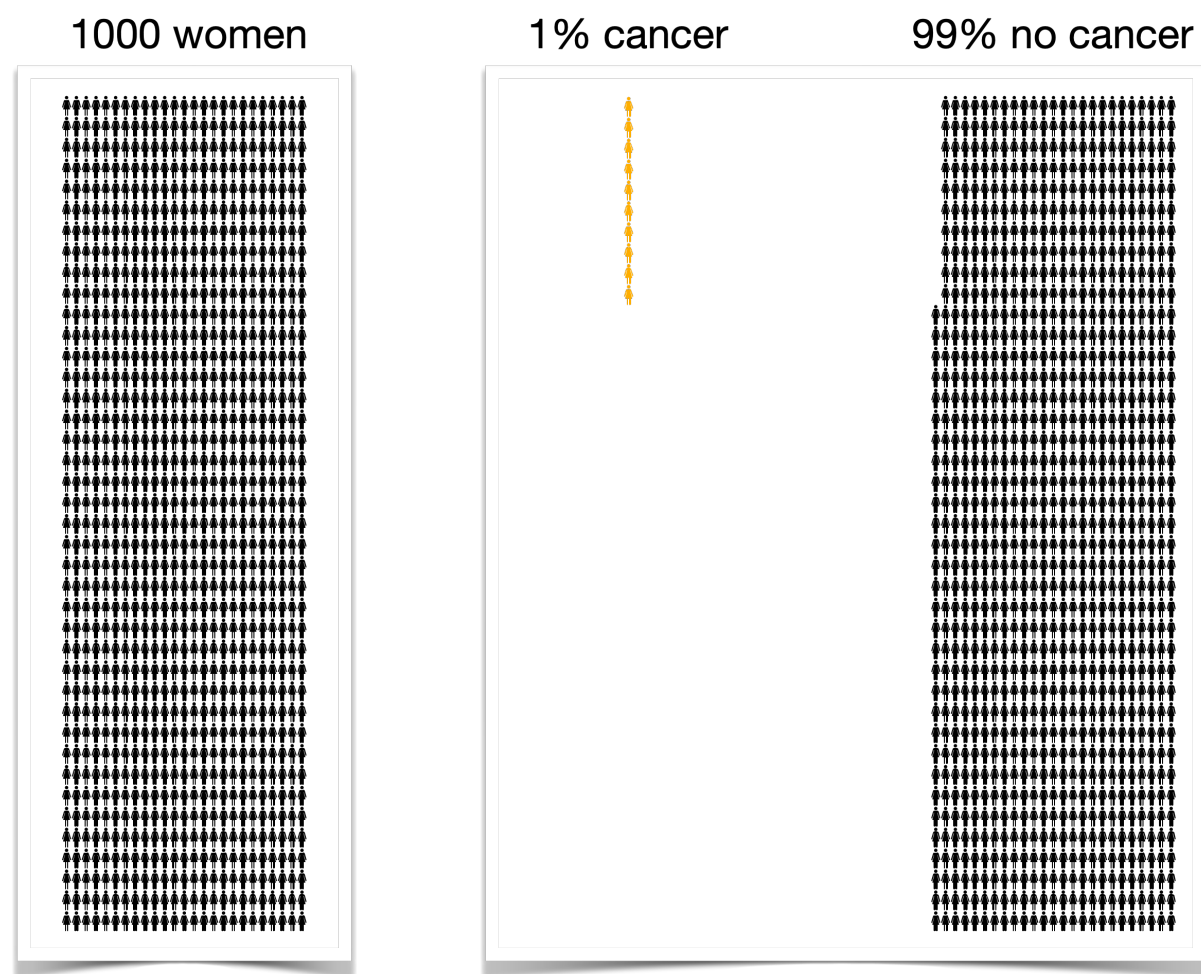
Overview:

1. Bayes Factor
2. Bayesian Information Criterion (BIC)
3. False discovery corrections
4. Bootstrapping for standard error and confidence
5. Shuffling for null distributions

1. Bayes factor

Let's refresh our Bayesian background and review **priors**, **posteriors**, and the **Bayes factor** for additional observations that can either lengthen or shorten the odds.

We start with a famous example from medical psychology and retrace the brilliant exposition by youtuber [Grant Sanderson \(3blue1brown\)](#). Specifically, we consider the incidence of breast cancer in middle-aged women and the correct interpretation of screening tests.



The incidence of breast cancer in 50 year-old women **without symptoms** is approximately 1%, or ten women in one thousand. Accordingly, the **prior odds** that any member of this population does or does not have cancer are

$$\text{Prior Odds} = \frac{P(\text{cancer})}{P(\text{no cancer})} = \frac{1}{99}$$

Sensitivity and specificity of screening

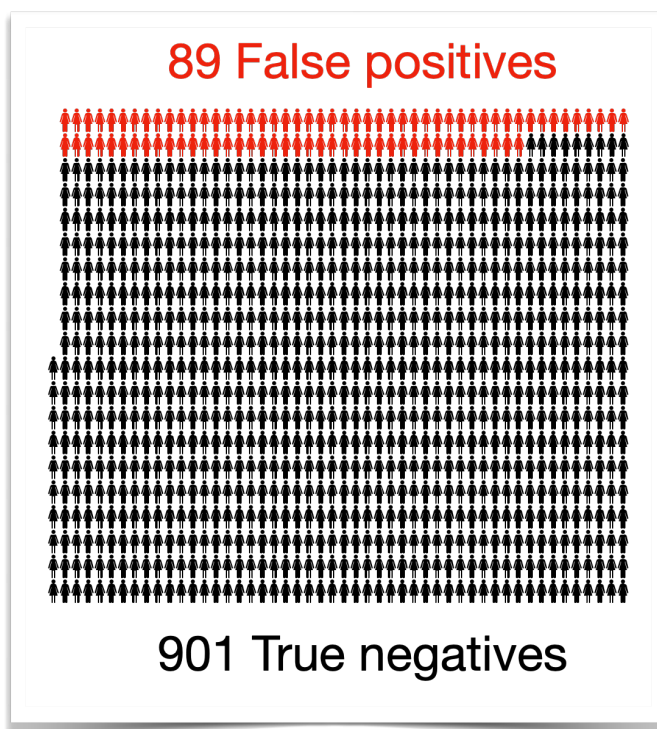
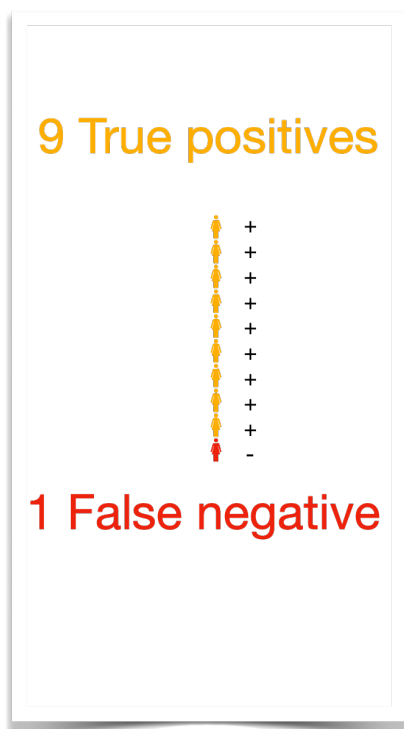
The **sensitivity** and the **false negative rate** (FNR) of a screening test are defined as

conditional probabilities for the **cancer-positive** subpopulation.

$$\text{Sensitivity} = P(+|\text{cancer}) = \frac{9}{10} = 90\%, \quad \text{FNR} = P(-|\text{cancer}) = \frac{1}{10} =$$

1% cancer

99% no cancer



Sensitivity = 90%

Specificity = 91%

In contrast, the **specificity** and the **false positive rate (FPR)** of a screening test are defined as conditional probabilities for the **cancer-negative** subpopulation.

$$\text{Specificity} = P(-|\text{no cancer}) = \frac{901}{990} \approx 91\%, \quad \text{FPR} = P(+|\text{no cancer}) =$$

Interpreting a screening result

So how should we interpret the result of a screening test?

The **posterior odds** are defined as the ratio of conditional probabilities, times the ratio of **prior odds**.

For a **positive** test, we find

$$\text{Posterior Odds} \equiv \frac{P(\text{cancer}|+)}{P(\text{no cancer}|+)} = \frac{P(+|\text{cancer})}{P(+|\text{no cancer})} \cdot \frac{P(\text{cancer})}{P(\text{no cancer})} = \frac{90\%}{9\%}.$$

In other words, a positive tests shortens our odds from 1/99 before the test to 1/10 after the test.

For a **negative** test, we find

$$\text{Posterior Odds} \equiv \frac{P(\text{cancer}|-)}{P(\text{no cancer}|-)} = \frac{P(-|\text{cancer})}{P(-|\text{no cancer})} \cdot \frac{P(\text{cancer})}{P(\text{no cancer})} = \frac{10\%}{91\%}.$$

so that a negative tests lengthens our odds from 1/99 before the test to 1/900 after the test.

Probability tables

The above equates the ratios of conditional probabilities (for the **same condition!**) to the ratios of joint probabilities. Why should this be case?

As always, the comprehensive picture is provided by the probability tables for conditional and joint probabilities.

In our example, the **conditional probabilities of test results** were given:

$P(\text{cancer})$	$P(\text{no cancer})$
0.01	0.99
$P(+ \text{cancer})$	$P(+ \text{no cancer})$
0.9	0.09
$P(- \text{cancer})$	$P(- \text{no cancer})$
0.1	0.91

Multiplying conditionals with priors (columnwise), we obtain the **joint probabilities** and the **marginal probabilities** (rowwise).

$P(\text{cancer})$	$P(\text{no cancer})$	
0.01	0.99	
$P(+, \text{cancer})$	$P(+, \text{no cancer})$	$P(+)$
0.009	0.0891	0.0981
$P(-, \text{cancer})$	$P(-, \text{no cancer})$	$P(-)$
0.001	0.9009	0.9019

Dividing joints by marginals (rowwise), we obtain the **conditional probabilities of cancer**:

$P(\text{cancer})$	$P(\text{no cancer})$	
0.01	0.99	
$P(\text{cancer} +)$	$P(\text{no cancer} +)$	$P(+)$
0.0917	0.9083	0.0981
$P(\text{cancer} -)$	$P(\text{no cancer} -)$	$P(-)$
0.0011	0.9989	0.9019

Now we can understand the equivalence of the ratios of conditional probabilities (same condition!) and joint probabilities:

$$\frac{P(\text{cancer}|+) \cdot P(+)}{P(\text{no cancer}|+) \cdot P(+)} = \frac{P(\text{cancer}, +)}{P(\text{no cancer}, +)},$$

$$\frac{P(\text{cancer}|-) \cdot P(-)}{P(\text{no cancer}|-) \cdot P(-)} =$$

$$\frac{P(+|\text{cancer}) \cdot P(\text{cancer})}{P(-|\text{cancer}) \cdot P(\text{cancer})} = \frac{P(+, \text{cancer})}{P(-, \text{cancer})},$$

$$\frac{P(+|\text{no cancer}) \cdot P(\text{no cancer})}{P(-|\text{no cancer}) \cdot P(\text{no cancer})} =$$

The twins of the physicist

Another example of Bayesian reasoning is the following story: thanks to sonograms, a physicist found out she was going to have twin boys. "What is the probability my twins will be *Identical*, rather than *Fraternal*?" she asked. The doctor answered that one-third of twin births were *Identicals*, and two-thirds *Fraternals*, so prior probabilities of 1/3 and 2/3, respectively.

In addition, we know from the sonogram that the two children will be of the same sex (boys or girls is immaterial here). The crucial background fact is that *Identical* twins are always same-sex while *Fraternal*s have probability 0.5 of having the same or different sex. In other words, the conditional probabilities of same sex, given *Identical* or *Fraternal* twins, are 1 and 1/2, respectively.

Collecting this information in standard notation, we have

$$\begin{aligned} \pi_{\text{frat}} &= 2/3, & \pi_{\text{ident}} &= 1/3 \\ \Pr(\text{same sex}|\text{frat}) &= 1/2, & \Pr(\text{same sex}|\text{ident}) &= 1 \end{aligned}$$

Combining this into Bayes fractions, we have

$$\frac{\Pr(\text{frat}|\text{samesex})}{\Pr(\text{ident}|\text{samesex})} = \frac{\Pr(\text{samesex}|\text{frat})}{\Pr(\text{samesex}|\text{ident})} \cdot \frac{\pi_{\text{frat}}}{\pi_{\text{ident}}} = \frac{1/2}{1} \cdot \frac{2/3}{1/3} = 1$$

In other words, the prior odds **against** *Identical* twins were levelled out by the same-sex sonogram, which **favours** *Identical* twins.

Probability of aliens (optional)

Let p be the prior probability that a star system is inhabited by a sentient species. If we survey N star systems (including our own), the binomial probability of observing exactly n inhabited systems is

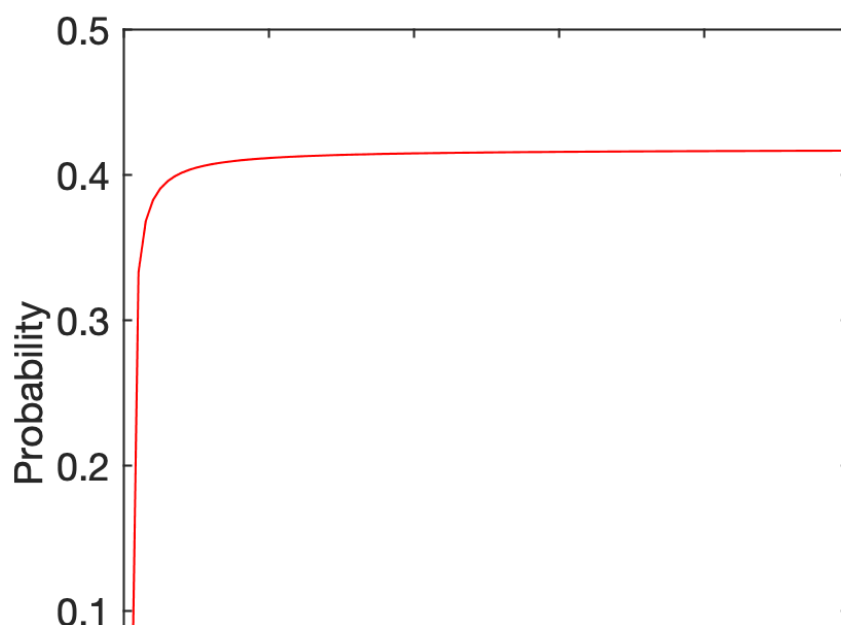
$$P_n(p, N) = \binom{N}{n} p^n (1-p)^{N-n}$$

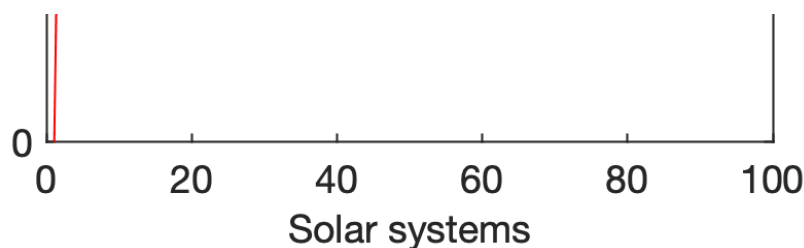
The conditional probability of two or more inhabited system, given that there already exists one such system (our own) is

$$P_{\text{alien}}(p, N) = \frac{1 - P_0(p, N) - P_1(p, N)}{1 - P_0(p, N)}$$

A conservative estimate for the prior probability is $p = \frac{1}{N}$, since our own system exists. Accordingly, we can estimate the probability of aliens, as a function of N , as follows

$$P_{\text{alien}}(N) = \frac{1 - P_0(1/N, N) - P_1(1/N, N)}{1 - P_0(1/N, N)}$$





This calculation shows that the probability of finding aliens asymptotes at 42% as the number of surveyed solar systems increases. Douglas Adams knew this all along.

Cautionary note: due to our limited knowledge, we assign the same prior probability to microbial life, multicellular macroscopic life, and sentient life. Surely these will turn out to be vastly different!

1. Bayesian Information Criterion (BIC)

To motivate this approach, let's for one moment return to our statistical utopia where everything is known or calculable.

Consider an independent normal sample

$$\mathbf{x} = (x_1, x_2, \dots, x_n), \quad x_i \sim \mathcal{N}(\mu, 1)$$

and the problem of distinguishing between two hypotheses or 'models'.

$$\mathcal{M}_0 : \mu = 0, \quad \mathcal{M}_1 : \mu \neq 0$$

As we inhabit a utopian scenario, we know the prior probabilities of both models

$$\Pr(\mathcal{M}_0) = \pi_0, \quad \Pr(\mathcal{M}_1) = \pi_1 = 1 - \pi_0$$

as well as the conditional prior densities of μ under both models

$$g_0(\mu) = g(\mu | \mathcal{M}_0), \quad g_1(\mu) = g(\mu | \mathcal{M}_1)$$

If $f_\mu(\mathbf{x})$ is the density of $\mathbf{x}$ given μ , the marginal distributions of $\mathbf{x}$ follow as

$$f_0(\mathbf{x}) = \int f_\mu(\mathbf{x}) g_0(\mu) d\mu, \quad f_1(\mathbf{x}) = \int f_\mu(\mathbf{x}) g_1(\mu) d\mu$$

From Bayes' theorem, we can now obtain the posterior probability ratio of models, given our observations $\mathbf{x}$,

$$\frac{\pi_1(\mathbf{x})}{\pi_0(\mathbf{x})} = \frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} \cdot \frac{\pi_1}{\pi_0}$$

as the product of the Bayes factor $B(\mathbf{x}) = \frac{f_1(\mathbf{x})}{f_0(\mathbf{x})}$ and the prior probability ratio $\frac{\pi_1}{\pi_0}$

Bayesian information criterion (BIC)

The BIC is a heuristic that compares quality of model fit and model complexity. Specifically

$$\text{BIC} = -2 \log(\hat{\mathcal{L}}) + p \log(n)$$

where $\hat{\mathcal{L}} = f_1(\mathbf{x})$ is the likelihood of the observed data $\mathbf{x}$ under model $\mathcal{M}_1$, p is the number of model parameters, and n is the number of data points.

A **lower BIC** indicates a **better model**.

✓ BIC for normally distributed observations

For normally distributed observation errors, the likelihood of observations is

$$f_1(\mathbf{x}) = \frac{1}{(2\pi \langle \sigma_e^2 \rangle)^{n/2}} \exp \left(-\frac{1}{2 \langle \sigma_e^2 \rangle} \sum_{i=1}^n (x_i - \mu)^2 \right)$$

where error variance is estimated from the **residual sum of squares, RSS**

$$\langle \sigma_e^2 \rangle = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

so that the log likelihood becomes

$$\begin{aligned} \log f_1(\mathbf{x}) &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \langle \sigma_e^2 \rangle - \frac{1}{2 \langle \sigma_e^2 \rangle} \sum_{i=1}^n (x_i - \mu)^2 = \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \langle \sigma_e^2 \rangle - \frac{n}{2} \end{aligned}$$

and the BIC becomes

$$\begin{aligned} \text{BIC} &= -2 \log f_1(\mathbf{x}) + p \log n = \\ &= n \log(2\pi) + n \log \langle \sigma_e^2 \rangle + n + p \log n \propto \end{aligned}$$

$$\propto n \log \langle \sigma_e^2 \rangle + p \log n$$

The constant terms $n \log(2\pi) + n$ can be ignored for purposes of model comparison.

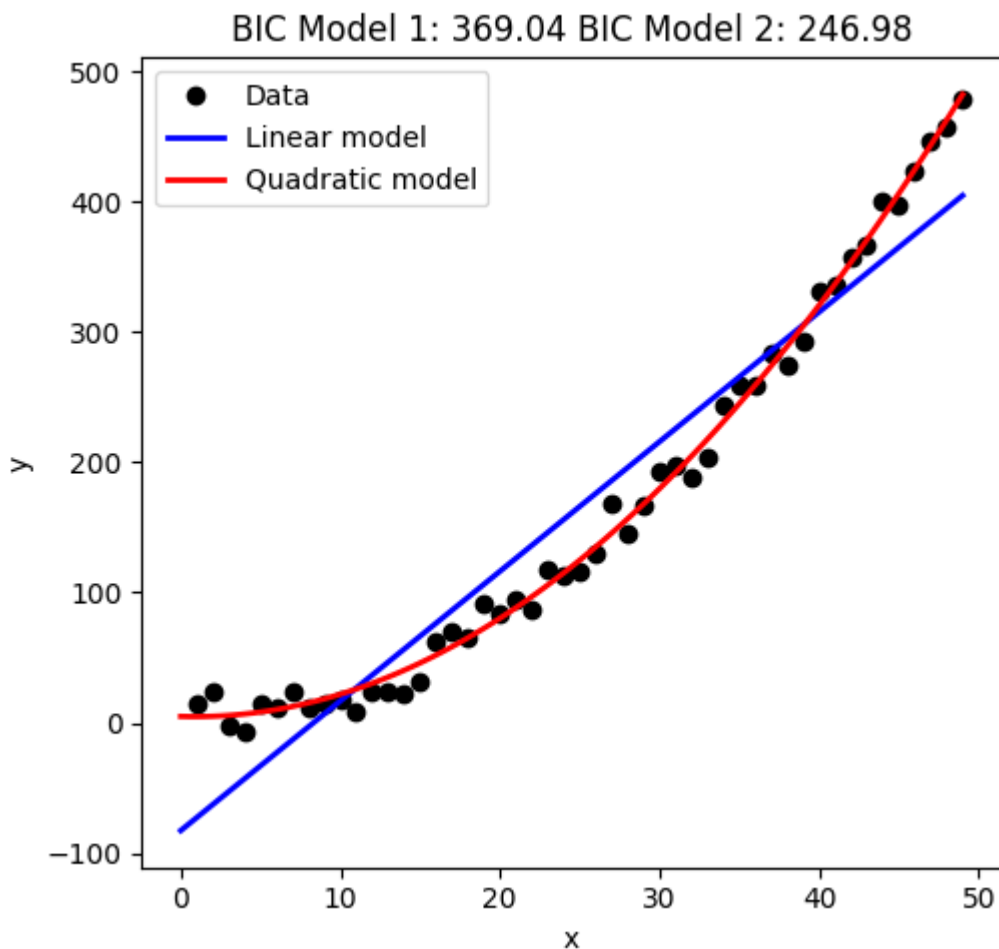
BIC with normally distributed observations

Show code

n 50

sigma_e 9

Bayes factors: 1.0256635813465204e+53 4.0607186941025504e-57



```
plot_BIC_with_normal_data
def plot_BIC_with_normal_data(n=20, sigma_e=2)

<no docstring>
```

Bayes factor and BIC

The BIC lets us compare models in terms of a Bayes factor, if we assume

$$P(\mathcal{M}_1|\mathbf{x}) = e^{-BIC(\mathcal{M}_1)}, \quad P(\mathcal{M}_0|\mathbf{x}) = e^{-BIC(\mathcal{M}_0)}$$

$$\log B_{BIC}(\mathbf{x}) = BIC(\mathcal{M}_0) - BIC(\mathcal{M}_1) = -2 \log \frac{f_0(\mathbf{x})}{f_1(\mathbf{x})} + (p_0 - p_1) \log(n)$$

$$B_{BIC}(\mathbf{x}) = \frac{P(\mathcal{M}_1|\mathbf{x})}{P(\mathcal{M}_0|\mathbf{x})} = \exp[BIC(\mathcal{M}_0) - BIC(\mathcal{M}_1)] = \left[\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} \right]^2 \cdot \frac{n^{p_0}}{n^{p_1}}$$

that describes how our observations $\mathbf{x}$ change the odds in favour of one model or another.

In the coded toy example above, the BIC of Model 1 is ~ 80 , whereas that of Model 2 is ~ 30 , so that the odds change by a factor of 10^{18} in favour of Model 2:

$$B_{BIC} \approx e^{-30+80} \approx 10^{18}$$

✓ BIC with Poisson distributed observations

For Poisson distributed observations $\mathbf{y}$ with expectation $\boldsymbol{\mu} = \langle \mathbf{y} \rangle$, the natural parameter is $\boldsymbol{\lambda} = \log \boldsymbol{\mu}$, which we fit as

$$\boldsymbol{\lambda}(\boldsymbol{\alpha}) = \log \boldsymbol{\mu}(\boldsymbol{\alpha}) = \mathbf{X} \cdot \boldsymbol{\alpha}$$

The Poisson distribution is

$$f_{\mu}(y) = \frac{\mu^y e^{-\mu}}{y!}, \quad \log f_{\mu}(y) = y \log \mu - \mu - \text{const} = y\lambda - e^{\lambda} - \text{const}$$

so that the (parameter-dependent part of the) log likelihood of observations given model predictions $\boldsymbol{\mu}(\boldsymbol{\alpha})$ is

$$\log f_{\boldsymbol{\mu}}(\mathbf{y}) = \sum_i \log f_{\mu_i}(y_i) = \sum_i [y_i \log \mu_i - \mu_i] = \sum_i [y_i \lambda_i - e^{\lambda_i}]$$

The BIC then becomes

$$BIC_{Poisson} = -2 \log f_{\boldsymbol{\mu}}(\mathbf{y}) + p \log(n)$$

where p is the number of parameters in $\boldsymbol{\alpha}$ plus 1 (for error variance) and n is the number of observations.

The Bayes factor between two models $\mathcal{M}_1$ and $\mathcal{M}_2$ is

$$\frac{P(\mathcal{M}_1|\mathbf{x})}{P(\mathcal{M}_2|\mathbf{x})} = \exp[BIC(\mathcal{M}_2) - BIC(\mathcal{M}_1)] = \left[\frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \right]^2 \cdot \frac{n^{p_2}}{n^{p_1}}$$

and the logarithm is

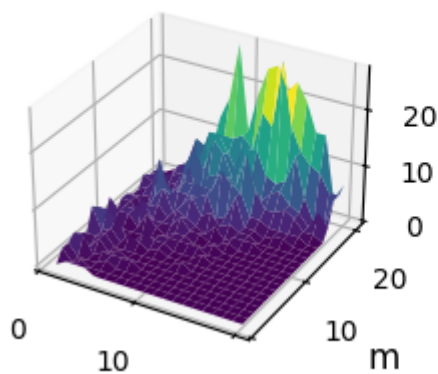
$$\begin{aligned}\log \left[\frac{P(\mathcal{M}_1|\mathbf{x})}{P(\mathcal{M}_2|\mathbf{x})} \right] &= 2 \log \left[\frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \right] + (p_2 - p_1) \log(n) = \\ &= 2 \sum_i y_i \left[\lambda_i^{(1)} - \lambda_i^{(2)} \right] - 2 \sum_i \left[e^{\lambda_i^{(1)}} - e^{\lambda_i^{(2)}} \right] + (p_2 - p_1) \log(n)\end{aligned}$$

BIC example for Poisson regression

Show code

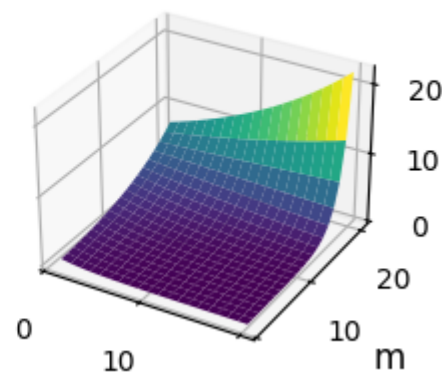
-714.7021431317858 -997.8471338196173

Observations

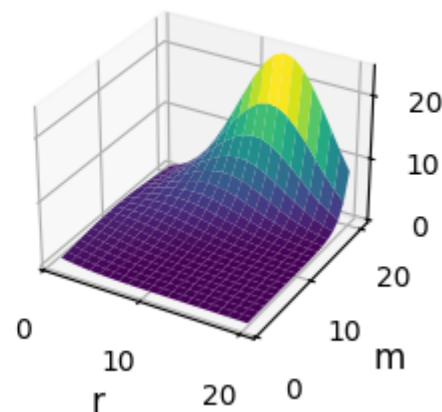
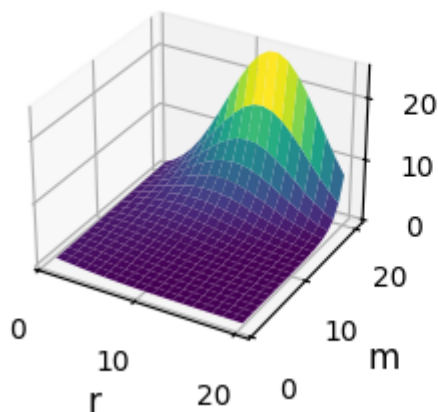


M1, BIC diff: -714.70

Model M0



M2, BF: -997.85



3. False discovery corrections

Many modern techniques with large-scale testing pose the statistical problem of false (accidental) positives. Performing 100 separate tests at significance level $\alpha = 0.05$, will **falsely discover** approximately five "significant" purely by chance.

Efron & Hastie (Chapter 15) present an example from a **prostate cancer** study with microarrays involving 102 men: 52 prostate cancer patients and 50 normal controls. For

microarrays, involving 102 men, 52 prostate cancer patients and 50 normal controls. For each man, expression levels were measured for 6033 genes, yielding a 6033×102 matrix of observations x_{ij} , with i indexing genes and j indexing men.

For each gene, a two-sample statistic compared expression levels patient and control groups. Under the null hypothesis H_0 that expression levels come from the same normal distribution, the t_i for gene i will follow a standard Student t distribution with 100 degrees of freedom, t_{100} . Transformation with the CDF of t_{100} and the inverse normal PDF produces standard normally distributed observations, z_i , under the null hypothesis:

$$z_i = \Phi^{-1} [F_{100}(t_i)]$$

The illustration below compares the observed distribution (histogram) with the null hypothesis (red curve), a standard normal distribution, scaled to counts:

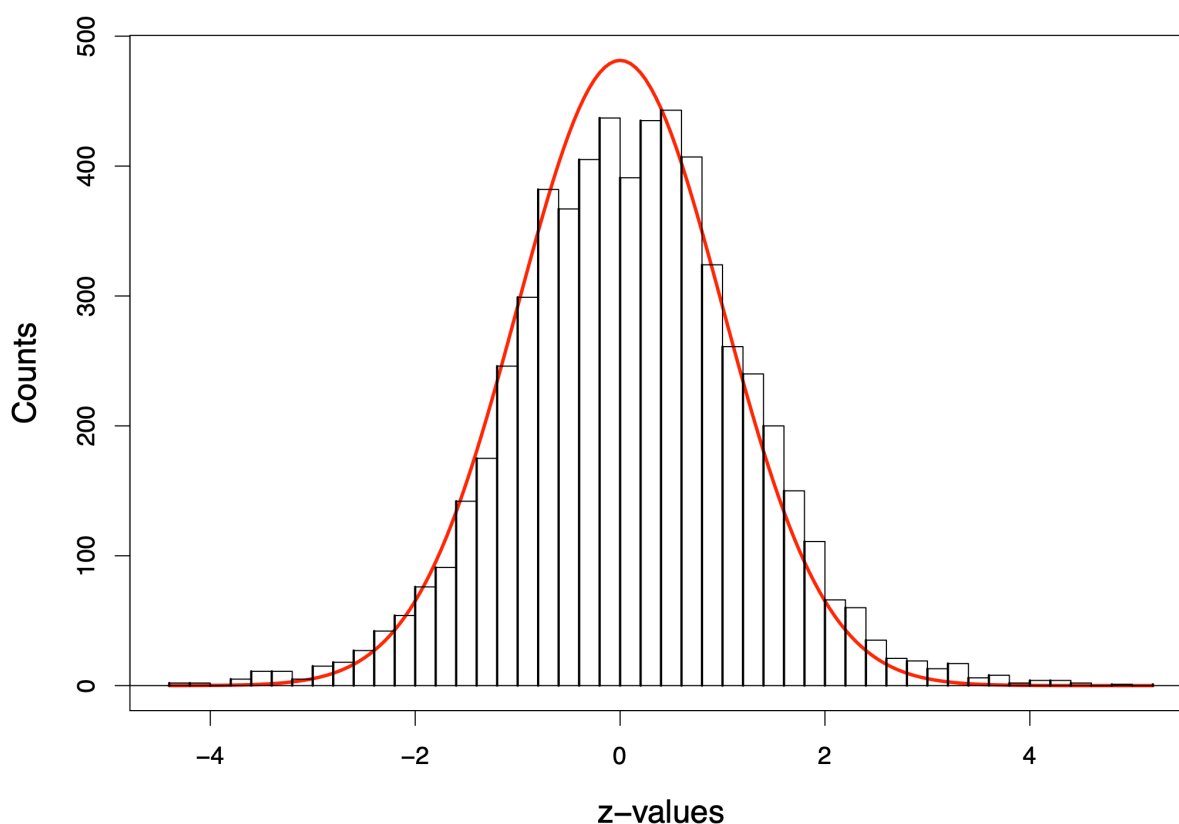


Figure 15.1 Histogram of $N = 6033$ z -values, one for each gene in the prostate cancer study. If all genes were null (15.3) the histogram would track the red curve. For which genes can we reject the null hypothesis?

So the question is, are some genes significantly over- or under-expressed?

Bonferroni correction

The classical Bonferroni correction lowers the threshold of significance from α for an individual test to α/N for N simultaneous tests. In other words, the null hypothesis is

rejected only when the p -value of some test is $p < \alpha/N$.

In the case of the prostate cancer study, with $N = 6033$, the Bonferroni-corrected threshold for significance would be

$$\alpha_{corr} = \frac{\alpha}{N} = \frac{0.05}{6033} = 8.3 \times 10^{-6}$$

This standard of evidence is quite stringent and only four genes in the cancer study pass this test.

Benjamini-Hochberg correction

The current standard is more liberal and was proposed by [Benjamini & Hochberg \(2019\)](#), whose paper has been cited 115,940 times to date!

If the true proportion of non-null hypotheses is $\frac{N_1}{N_0+N_1}$ and our statistical decision rule $\mathcal{D}$ yields a proportion of $\frac{R}{R+N}$, the the **false-discovery rate (FDR)** would be

$$\text{FDR}(\mathcal{D}) = \frac{a}{R}$$

Decision

		Null	Non-Null	
Actual	Null	$N_0 - a$	a	N_0
	Non-Null	$N_1 - b$	b	N_1
		$N - R$	R	N

We are seeking a decision rule that keeps the FDR below some chosen level, q

$$\text{FDR}(\mathcal{D}) < q$$

$$1 - \Phi(z_{(i)}) \leq q$$

The procedure for doing this is the order the observed p-values from the largest to the smallest, and to reject the null hypothesis for all values with

$$p_i \leq \frac{i}{N} q$$

with a conventional choice of $q = 0.1$.

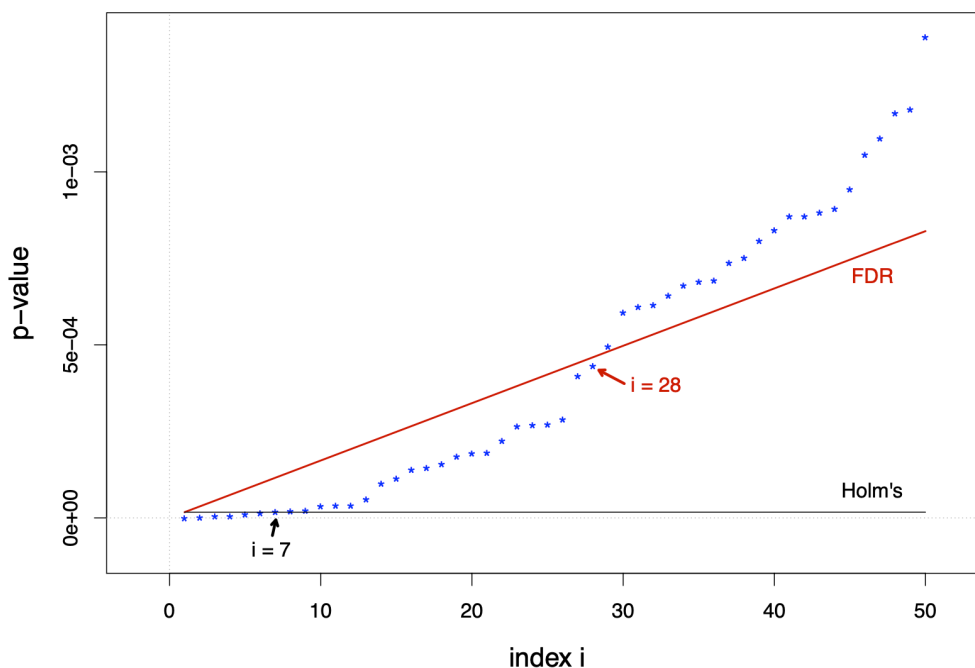


Figure 15.3 Ordered p -values $p_{(i)} = 1 - \Phi(z_{(i)})$ plotted versus i for the 50 largest z -values from the **prostate** data in Figure 15.1. The FDR control boundary (algorithm $\mathcal{D}_q$, $q = 0.1$)

In the cancer study, FDR rejects the null-hypothesis for 28 genes in the right tail of the distribution (over-expression).

Bayesian motivation for FDR

Benjamini & Hochberg derived their method in a different way (Martingale theory), but it can be easily motivated in a Bayesian way.

Assuming that our observations (gene expression levels) reflect a **mixed** population with a large fraction $\pi_0 \approx 1$ of expression levels distributed normally about mean zero

$$f_0(z) \sim \mathcal{N}(0, 1)$$

but a small fraction $\pi_1 = 1 - \pi_0$ distributed about some larger mean $\mu_1 > 0$

$$f_1(z) \sim \mathcal{N}(\mu_1, 1)$$

the mixed density will be

$$f_{mix}(z) = \pi_0 f_0(z) + \pi_1 f_1(z)$$

and the survivor fraction will be

$$S_{mix}(z_\theta) = \int_{z_\theta}^{\infty} f_{mix}(z) dz = \pi_0 S_0(z_\theta) + \pi_1 S_1(z_\theta)$$

If we rank our N observed z_i values, where i is the rank **from the largest** value, the false discovery rate at level of z_i is defined as

$$\text{FDR} = \frac{\pi_0 S_0(z_i)}{S_{mix}(z_i)}, \quad S_{mix}(z_i) \approx \frac{i}{N}$$

where $S_{mix}(z_i)$ is well approximated by $\frac{i}{N}$ for large N .

In other words, if we can tolerate a false discovery rate of q , we need to restrict the probability p_i of spurious discovery to

$$p_i = \pi_0 S_0(z_i) \leq S_{mix}(z_i) \cdot q \approx \frac{i}{N} \cdot q$$

Local false discovery rates

Let's apply this Bayesian viewpoint to the tail of our distributions and define a **local false discovery rate** for a small region $\mathcal{Z}_0$ centered on z_0

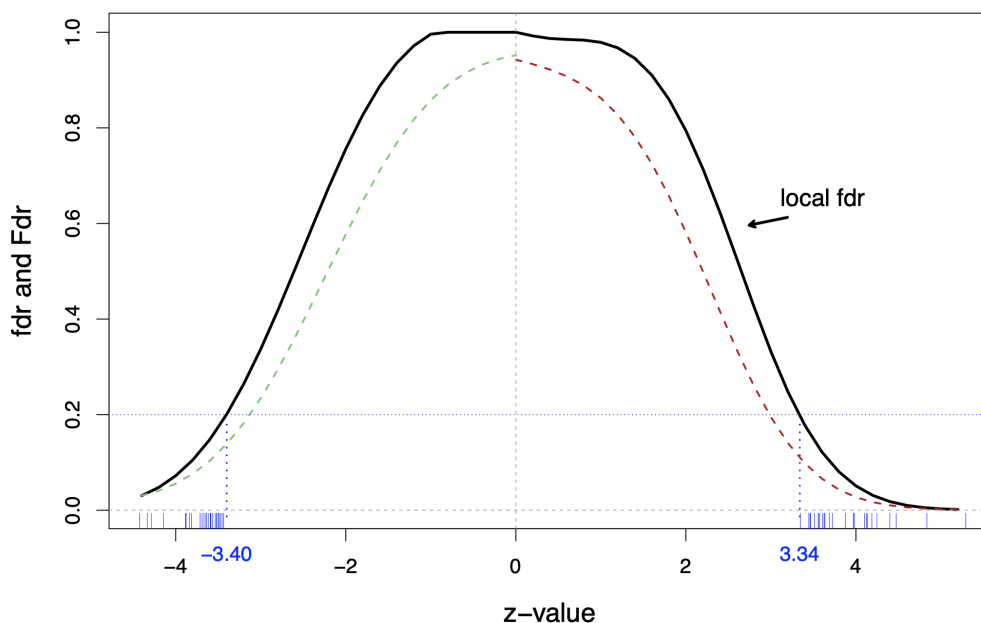
$$\text{fdr}(\mathcal{Z}_0) = \Pr(\text{null} | z_i \in \mathcal{Z}_0), \quad \mathcal{Z}_0 = \left[z_0 - \frac{d}{2}, z_0 + \frac{d}{2} \right]$$

Comparing the observed number of z_i in this region, $N(\mathcal{Z}_0)$, to the number expected from a standard normal distribution, $N_0(\mathcal{Z}_0)$, we can estimate

$$\text{fdr}(\mathcal{Z}_0) \approx \frac{N_0(\mathcal{Z}_0)}{N(\mathcal{Z}_0)} = \frac{N \pi f_0(z_0) d}{N(\mathcal{Z}_0)}$$

$$\text{fdr}(\mathcal{Z}_0) = \frac{N(\mathcal{Z}_0)}{N(\mathcal{Z}_0)} \approx \frac{N(\mathcal{Z}_0)}{N(\mathcal{Z}_0)}$$

Typically, these estimates are computed for different window centers z_0 and then smoothed by some polynomial fit. Once again, this approach works only when the number of hypothesis tests is quite large.



In our cancer study, 27 genes in the right tail and 25 genes in the left tail passed the threshold $\text{fdr} \leq q$ (the conventional threshold of "interesting" or worthy of further study), in the sense that $z_i \geq 3.34$ or $z_i \leq -3.40$.

4. Bootstrapping for standard error and confidence

Often we have a (smallish) set of observations $\mathbf{x} = (x_1, x_2, \dots, x_n)$ from some unknown probability distribution F

$$x_i \sim F \quad i = 1, 2, \dots, n$$

and have computed a (real-valued) statistic θ with some algorithm or procedure $s(\mathbf{x})$

$$\theta = s(\mathbf{x})$$

Note that this situation is quite general: the samples x_i can be univariate, bivariate, multivariate, the distribution is not restricted to any parametric family, and the algorithm $s(\mathbf{x})$ can be anything.

So is there any way to estimate a standard error for $\hat{\theta}$, in other words, estimate the variance we would expect if (counterfactually) we could repeat our experiment many times?

Suprisingly, there is a simple (albeit computationally intensive) family of solutions for this problem, which are known collectively as **bootstrapping**.

✓ Jackknife estimate

This is a versatile and handy tool for many situations, much like a pocket knife or jackknife.

Let's remove one observation, x_i , from our sample $\mathbf{x}$, and denote with $\mathbf{x}_{(i)}$ the smaller sample that results

$$\mathbf{x}_{(i)} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

and with $\theta_{(i)}$ the corresponding value of our statistic

$$\theta_{(i)} = s(\mathbf{x}_{(i)})$$

The Jackknife estimate for standard error is

$$se_{jack} = \left[\frac{n-1}{n} \sum_{i=1}^n (\theta_{(i)} - \theta_{(\cdot)})^2 \right]^{-1/2}, \quad \theta_{(\cdot)} = \frac{1}{n} \sum_{i=1}^n \theta_{(i)}$$

As a transparent example, consider the case when $\theta = s(x)$ is simply the mean of real-valued x_i

$$\theta = \bar{x} = \frac{1}{n} \sum_i x_i, \quad \theta_{(i)} = \frac{n\bar{x} - x_i}{n-1}$$

In this case, the average and the deviation from the average are

$$\theta_{(\cdot)} = \frac{1}{n} \sum_i \theta_{(i)} = \frac{n\bar{x}}{n-1} - \frac{\bar{x}}{n-1} = \bar{x}, \quad \theta_{(i)} - \theta_{(\cdot)} = \frac{n\bar{x} - x_i}{n-1} - \frac{\bar{x}(n-1)}{n-1}$$

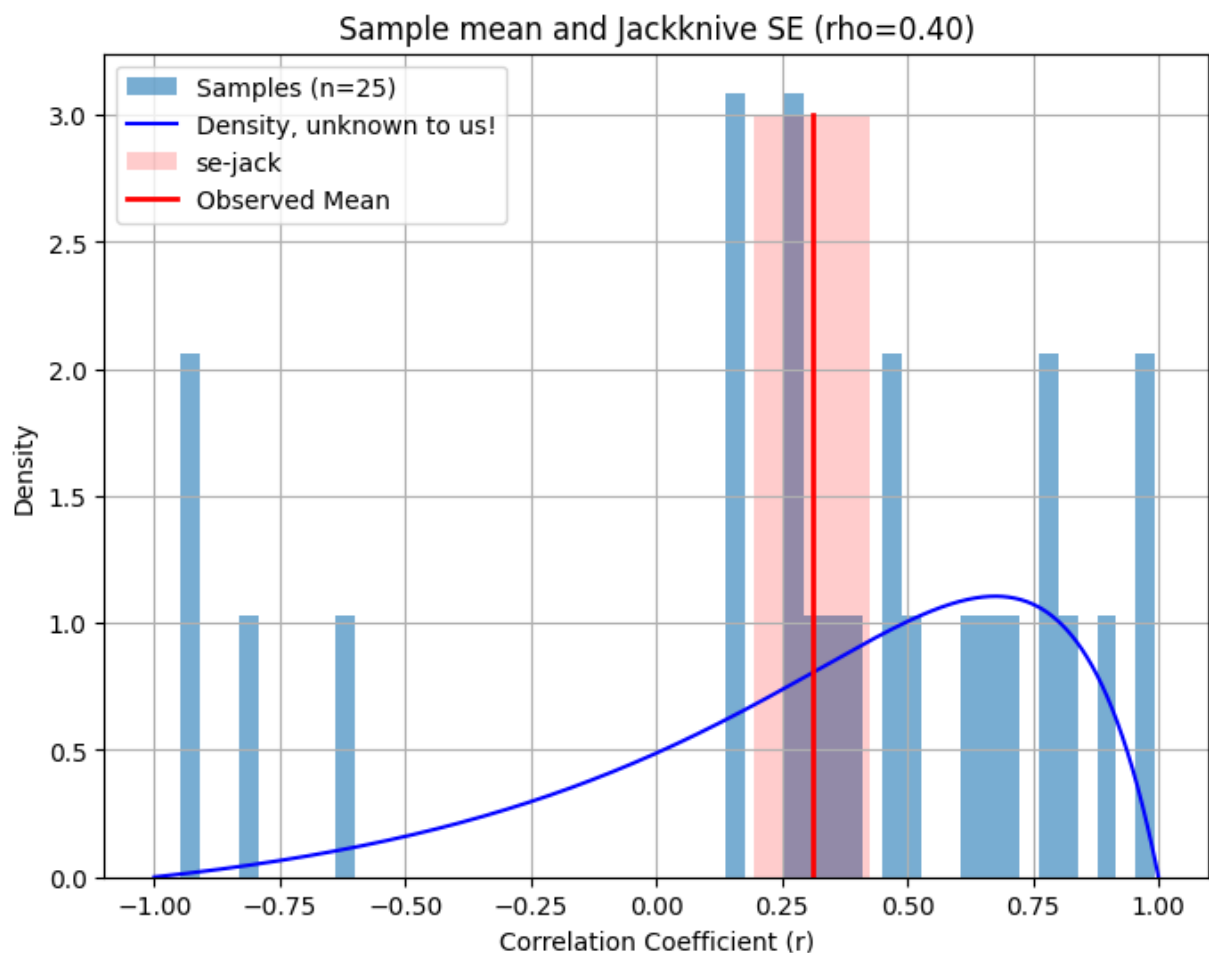
and the Jackknife estimate

$$se_{jack} = \left[\frac{n-1}{n} \sum_{i=1}^n (\theta_{(i)} - \theta_{(\cdot)})^2 \right]^{-1/2} = \left[\frac{1}{n(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{-1/2}$$

equals the classical standard error for normally distributed samples.

Jackknife example, sampling from Fisher's correlation density

Show code



✓ Nonparametric bootstrap

We stay with our general problem of a smallish set of observations $\mathbf{x} = (x_1, x_2, \dots, x_n)$ from some unknown probability distribution F

$$x_i \sim F \quad i = 1, 2, \dots, n$$

for which we have computed a (real-valued) statistic θ with some algorithm or procedure $s(\mathbf{x})$

$$\theta = s(\mathbf{x})$$

Bootstrapping estimates the unknown distribution F by generating from our on original sample many **bootstrap samples**

$$\mathbf{x}^* = (x_1^*, x_2^*, \dots, x_n^*)$$

by simply drawing each x_i^* randomly (with equal probability and with replacement) from $(x_1, x_2, \dots, x_n)$. From each bootstrap sample, we obtain a bootstrap replicate for our statistic θ

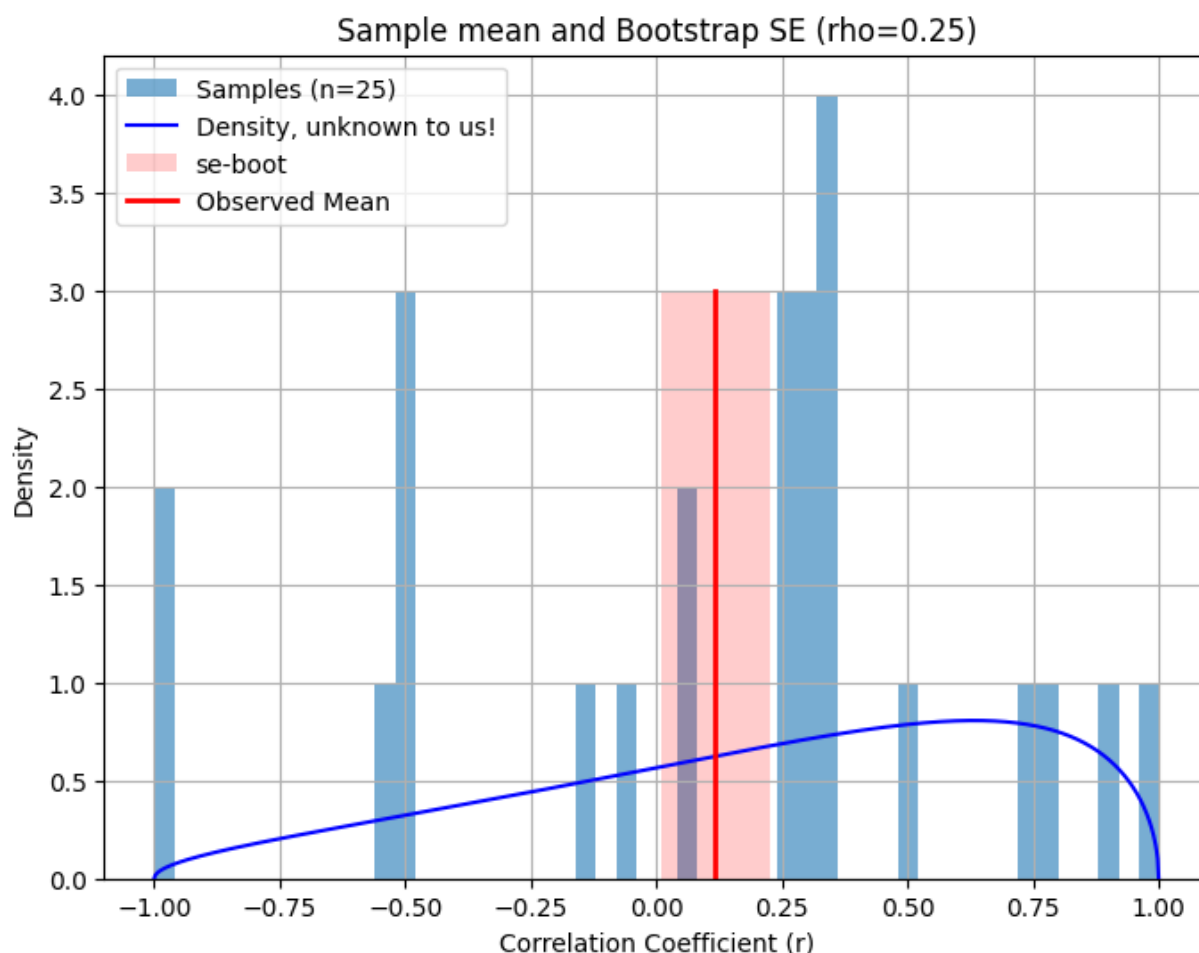
$$\theta_b = s(\mathbf{x}_b^*), \quad b = 1, 2, \dots, B$$

where b indexes the B repetitions of our bootstrap. The bootstrap standard error is then

$$se_{boot} = \left[\frac{1}{B-1} \sum_{b=1}^B (\theta_b - \theta_{(\cdot)})^2 \right]^{-1/2}, \quad \theta_{(\cdot)} = \frac{1}{B} \sum_{b=1}^B \theta_b$$

Bootstrap example, again with Fisher's correlation density

Show code



Resampling plans

If our sample size n is very small, the number of different possible Bootstrap samples is also small. For $n = 3$, we find that $N_{boot} = 10$ (see Figure below). In general, we have

$$N_{boot} = \binom{2n-1}{n}$$

$$N_{boot} = \binom{n}{1}$$

So as our sample gets larger, the number of possible bootstrap sample quickly becomes impractical: 92, 378 for $n = 10$ and 6.9×10^{10} for $n = 20$. So how can we ensure that a smallish number of bootstrap samples is unbiased representation of the full ensemble?

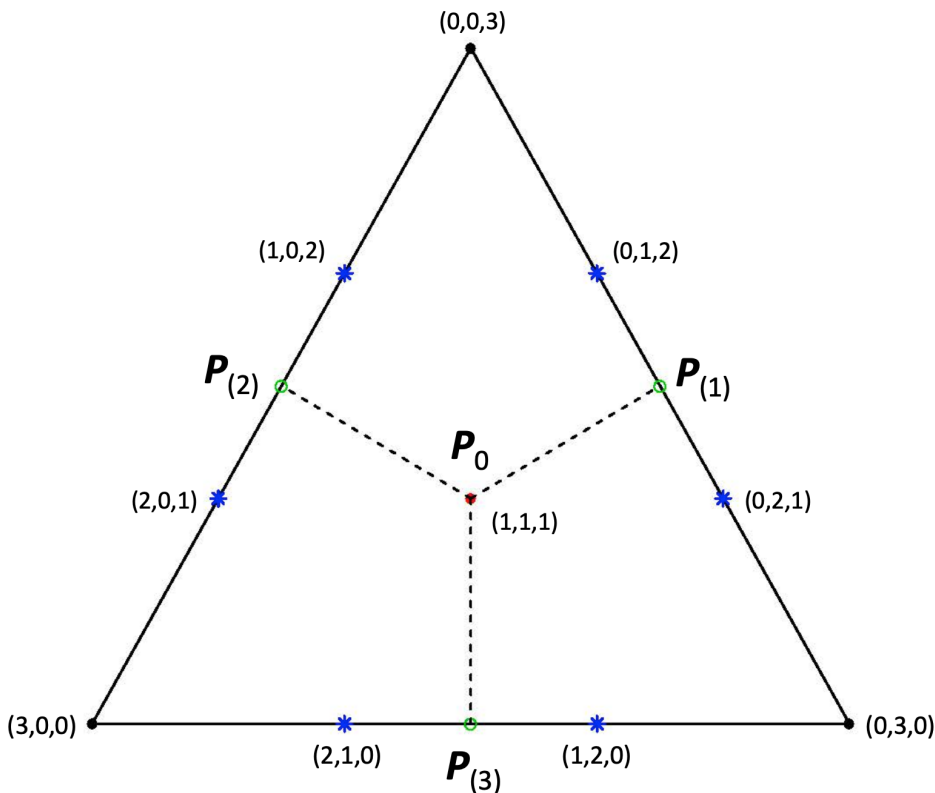


Figure 10.3 Resampling simplex for sample size $n = 3$. The center point is P_0 (10.26); the green circles are the jackknife points $P_{(i)}$ (10.28); triples indicate bootstrap resampling numbers (N_1, N_2, N_3) (10.29). The *bootstrap probabilities* are $6/27$ for P_0 , $1/27$ for each corner point, and $3/27$ for each of the six starred points.

We define a resampling vector as a set of non-negative weights summing to unity

$$\mathbf{P} = (P_1, P_2, \dots, P_n), \quad P_i \geq 0, \quad \sum_{i=1}^n P_i = 1$$

Mathematically, $\mathbf{P}$ is a point in the simplex $\mathcal{S}_n$, as illustrated above.

We can now apply our statistic to a probabilistically resampled version of the original sample $\mathbf{x}$:

$$\theta^* = s(\mathbf{x}^*), \quad \mathbf{x}^* = (P_1 x_1, P_2 x_2, \dots, P_n x_n)$$

For example, when our statistic is the arithmetic mean, we have

$$\theta^* = \frac{1}{n} \sum_i P_i x_i$$

When we choose resampling vectors $\mathbf{P}$ at random, we can get accurate estimates of the ensemble mean from reasonably small numbers of resampling, $B = 200$.

✓ Confidence intervals

To obtain confidence intervals, we can generate a larger number of ($B = 5000 \rightarrow 20000$) of bootstrapped samples $\mathbf{x}^*$ and corresponding statistics θ^* .

Sorting these statistics from the smallest to the largest value, we can identify the $\alpha/2$ level confidence tails. The $B\alpha/2$ smallest values are the low tail, and the $B\alpha/2$ larger values are the high tail.

The code below uses $B = 5000$ and $\alpha = 0.05$ so that both low and high tail comprise 125 values each. The corresponding distribution bins are shown in red.

In this example, the distribution from which our samples are drawn is known analytically (Fisher's correlation density) and the sample values are correlation coefficients. In this particular case, we can use Fisher's z-transform

$$z = \frac{1}{2} \log \frac{1+r}{1-r} = \text{arctanh}(r)$$

to obtain transformed values that are (very nearly) normally distributed, as illustrated by the second plot. This allows us to check our bootstrapped confidence interval by comparing it to the confidence interval of the normal distribution. As expected, everything matches beautifully!

(The one flaw is that the routine supplied by Grok undersamples the peak region of the distribution. However, this concerns the part of the code that generates synthetic bootstrap samples from an analytical distribution. Can we fix this by generating authentic bootstrap samples?)

To summarize, you can obtain confidence intervals for any kind of sample distribution by bootstrapping 5000 (or more) times and sorting the resulting statistics from the smallest to the lowest.

This is seriously useful!

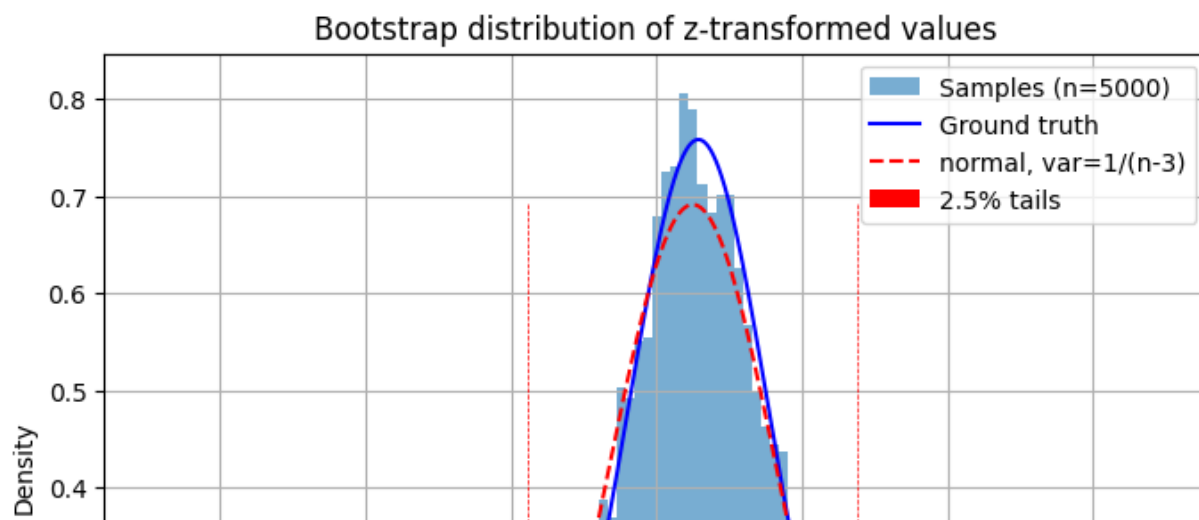
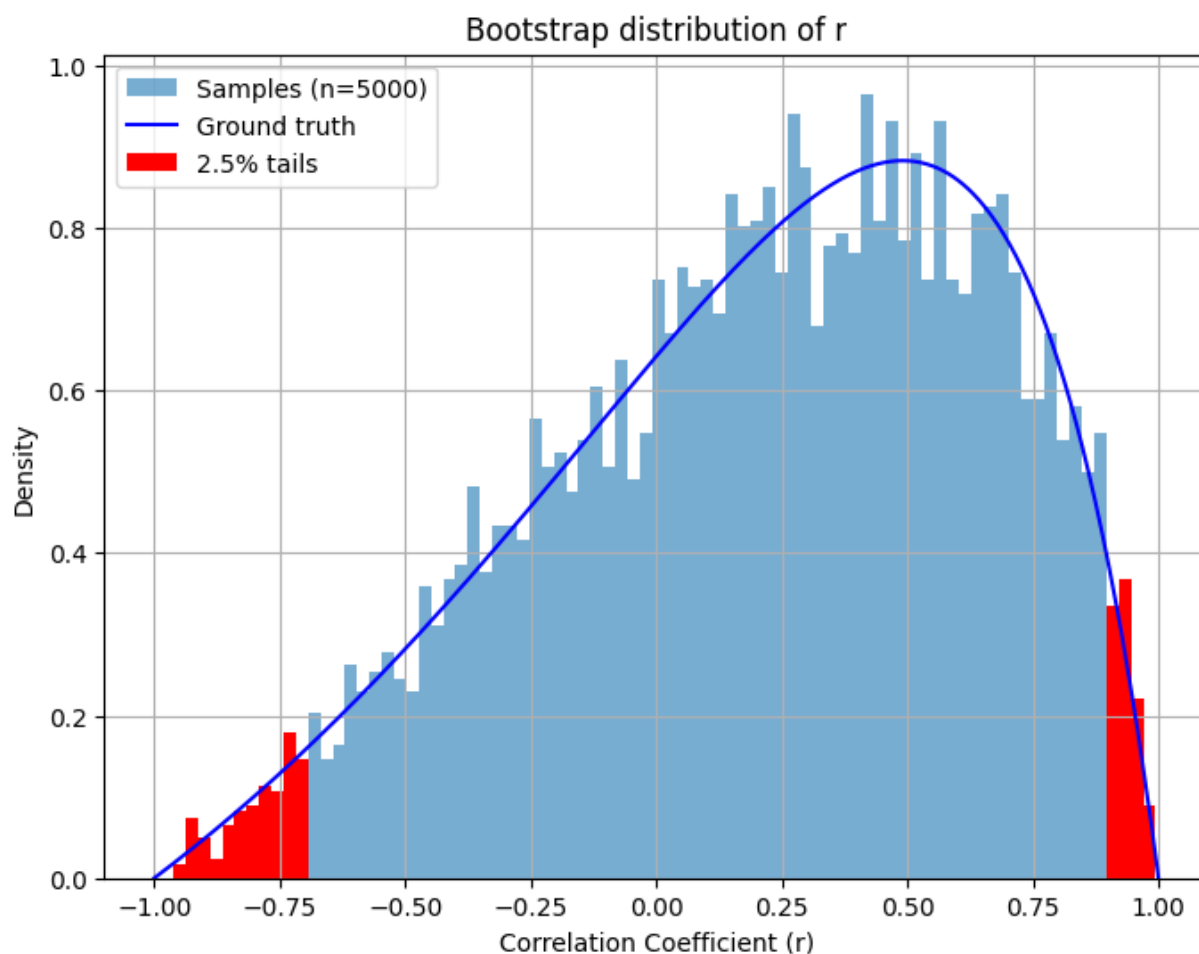
Confidence intervals from bootstrap distribution

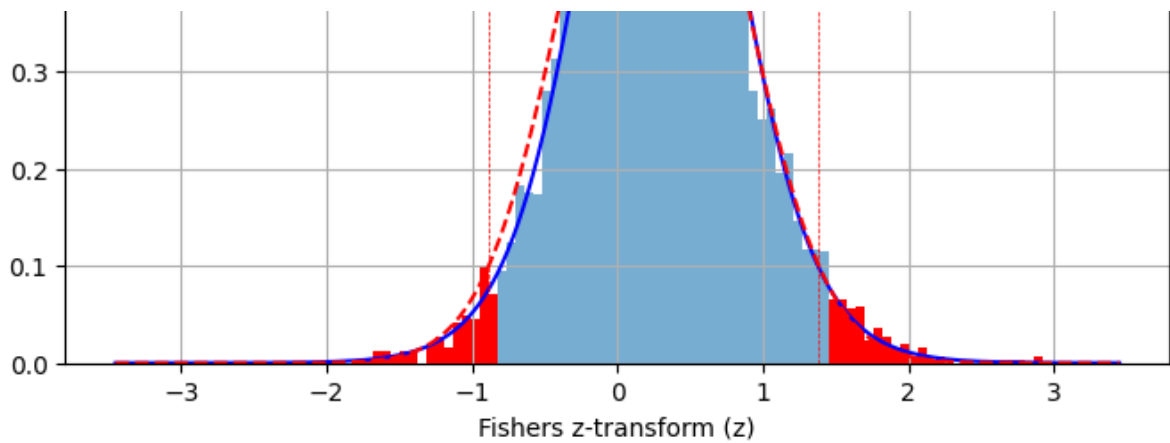
Show code

n 6
rho 0.25

```
<ipython-input-79-c37ecae50ab4>:134: RuntimeWarning: divide by zero
encountered in arctanh
```

```
z_values = np.arctanh( r_values)
```





```
plot_densities_of_r_and_z_values
def plot_densities_of_r_and_z_values(n=6, rho=0.25)

<no docstring>
```

Double-click (or enter) to edit

5. Shuffling for null distributions

One of the most useful computer-based techniques is the generation of null distributions.

When we design a specific statistical summary of complex data (multiunit recordings, EEG recordings, fMRI signals, etc), we rarely have extensive prior experience about the expected distribution under the null hypothesis. However, by selectively permuting those aspects of our data that pertain to our hypothesis, we can synthetically generate data under the null hypothesis. If we do this sufficiently often (10000 times), we can establish confidence limits under the null distribution, to which we can compare the statistic for the observed data.

Arrow-of-time asymmetry

Here I illustrate an application involving "arrow-of-time" asymmetry of fMRI signals. Arrow-of-time refers to the fact that many physical systems develop in a manner that is asymmetric in time. (A dropped glass shatters on the floor, but a shattered glass never hops up from the floor and flies into your hand.) Often this asymmetry is simply due to an observable time-lag between cause and effect. For example, neural activity in the retina, the lateral geniculate, and primary visual cortex propagates with an observable delay, so that retinal activity

consistently precedes lateral geniculate activity, which in turn consistently precedes primary visual cortex activity.

Consistent lags between two time-varying signals $x(t)$ and $y(t)$ are measured in terms of lagged correlations

$$\rho(\tau) = \langle x(t) \cdot y(t + \tau) \rangle, \quad \langle x(t) \rangle = \langle y(t) \rangle = 0, \quad \langle x^2(t) \rangle = \langle y^2(t) \rangle = 1$$

To quantify the degree asymmetry, we convert correlation coefficients to mutual information (using the standard formula for normally distributed signals)

$$I_m(\tau) = -\frac{1}{2} \log[1 - \rho^2(\tau)]$$

choose the sign of this mutual information such as to distinguish positive and negative correlations

$$I_{signed}(\tau) = I_m(\tau) \cdot \text{sign}[\rho(\tau)]$$

form the squared difference between signed mutual information at positive and negative lags

$$\text{FowRev}(\tau) = [I_{signed}(\tau) - I_{signed}(-\tau)]^2$$

and integrate this difference over a suitable window of time

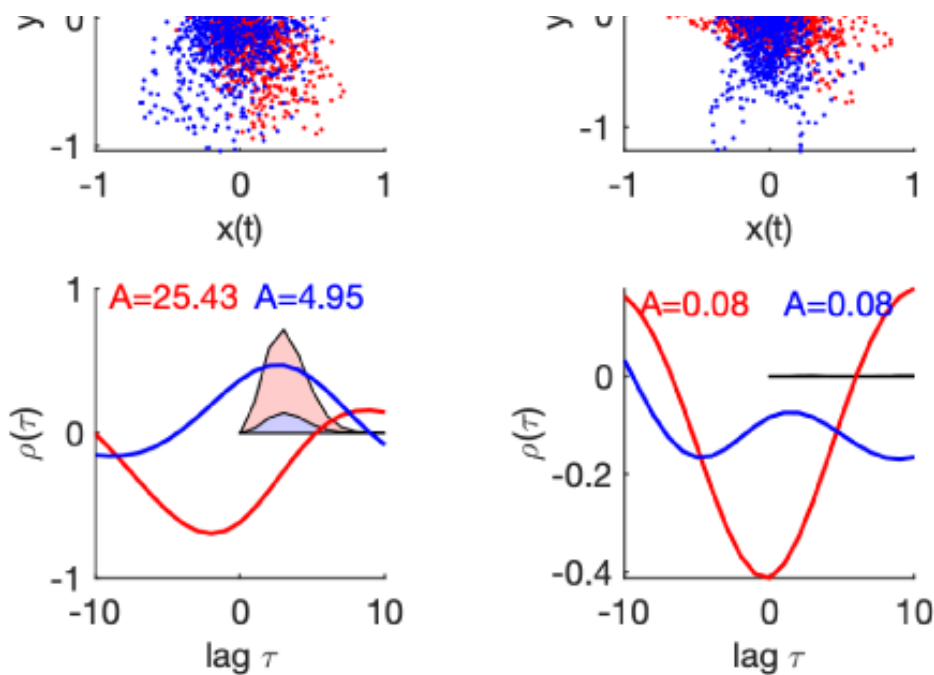
$$\text{Asym} = \int_0^T \text{FowRev}(\tau) d\tau$$

Examples for strong and weak asymmetry

In the illustration below, the upper panels show four signal pairs (1288 time points each), two strongly asymmetric and two weakly asymmetric.

The lower panels show the lagged correlations $\rho(\tau)$ and the squared difference $\text{FowRev}(\tau)$. Our measure of "arrow-of-time" asymmetry is the integral (area under the curve) integral.





✓ Null distribution

To generate a null distribution, the computation of "arrow-of-time" asymmetry values must be repeated many times for selectively randomized observations.

The relevant aspect for "arrow-of-time" asymmetry is the temporal alignment of the two time-varying signals, $x(t)$ and $y(t)$. To randomize this alignment, we can shift one signal by some random time T

$$y(t) \rightarrow y(t - T_{rand})$$

However, we should be careful to avoid the interval $-T \leq \tau \leq T$ for which we compute asymmetry, so that we do not accidentally shift the signals back into alignment.

The code below uses

$$2T \leq T_{rand} \leq 4T$$

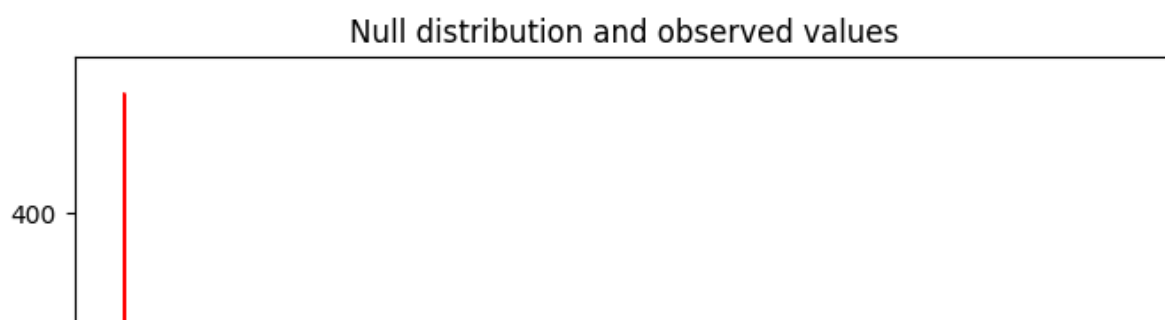
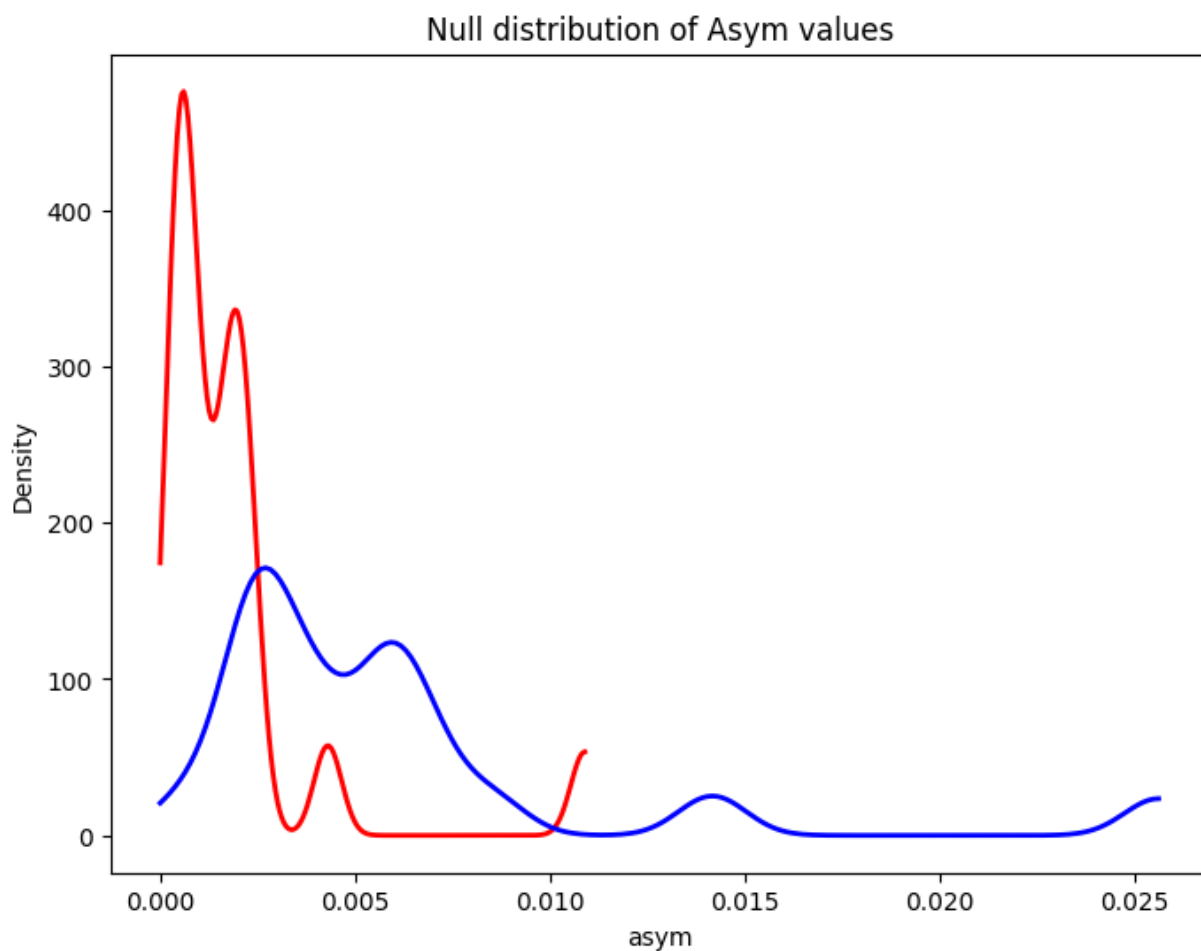
but any other interval will produce comparable results. The distribution of shuffled values is very close to zero, making our observed values **very highly significant**.

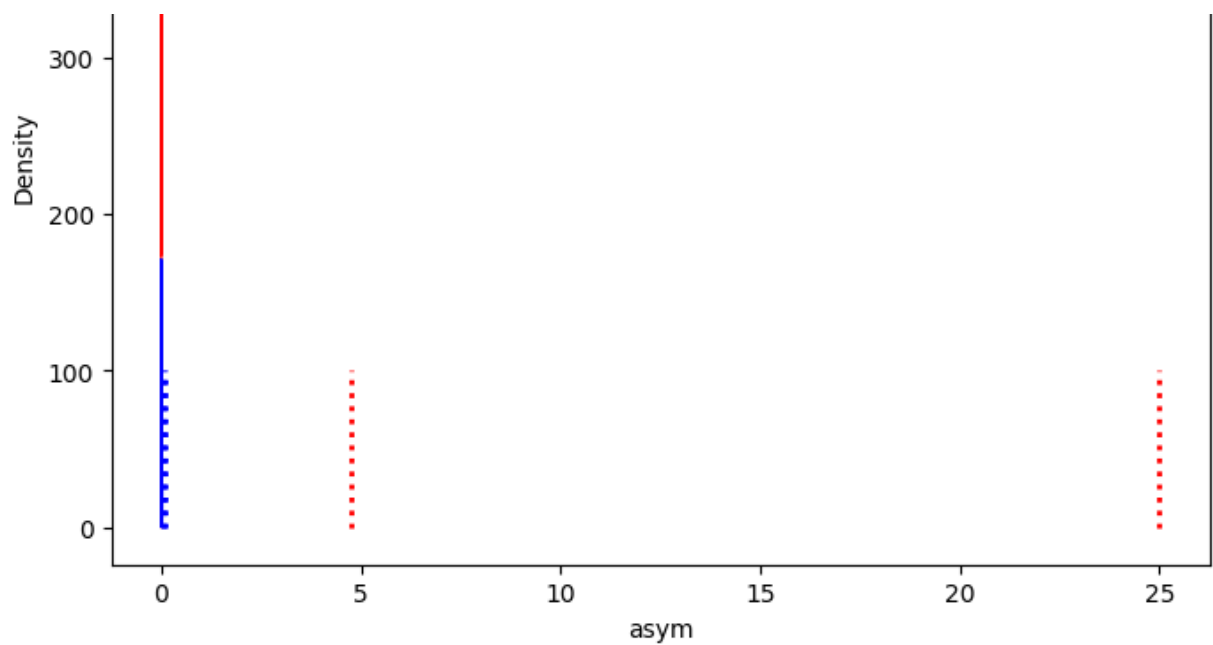
Null distribution for arrow-of-time asymmetry

Show code

Drive already mounted at /content/drive; to attempt to forcibly remount,

```
Variables in .mat file: dict_keys(['__header__', '__version__', '__global__'])
Observed asymmetry values: [25.00782867] [4.73800557] [0.07980689] [0.0720]
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
```



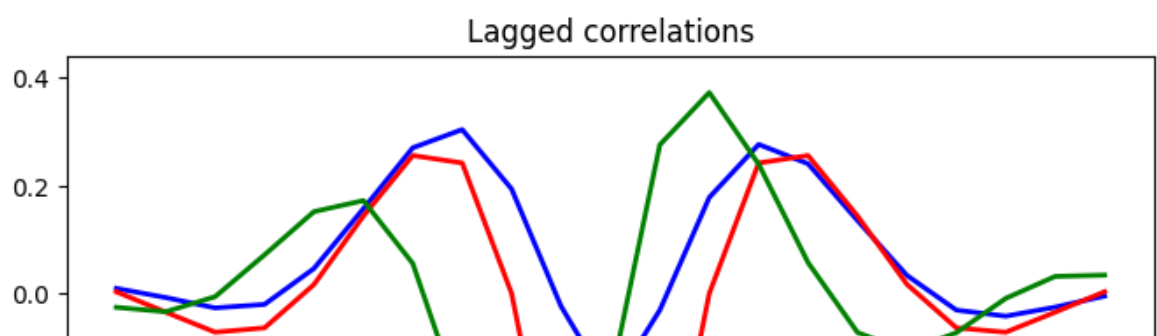


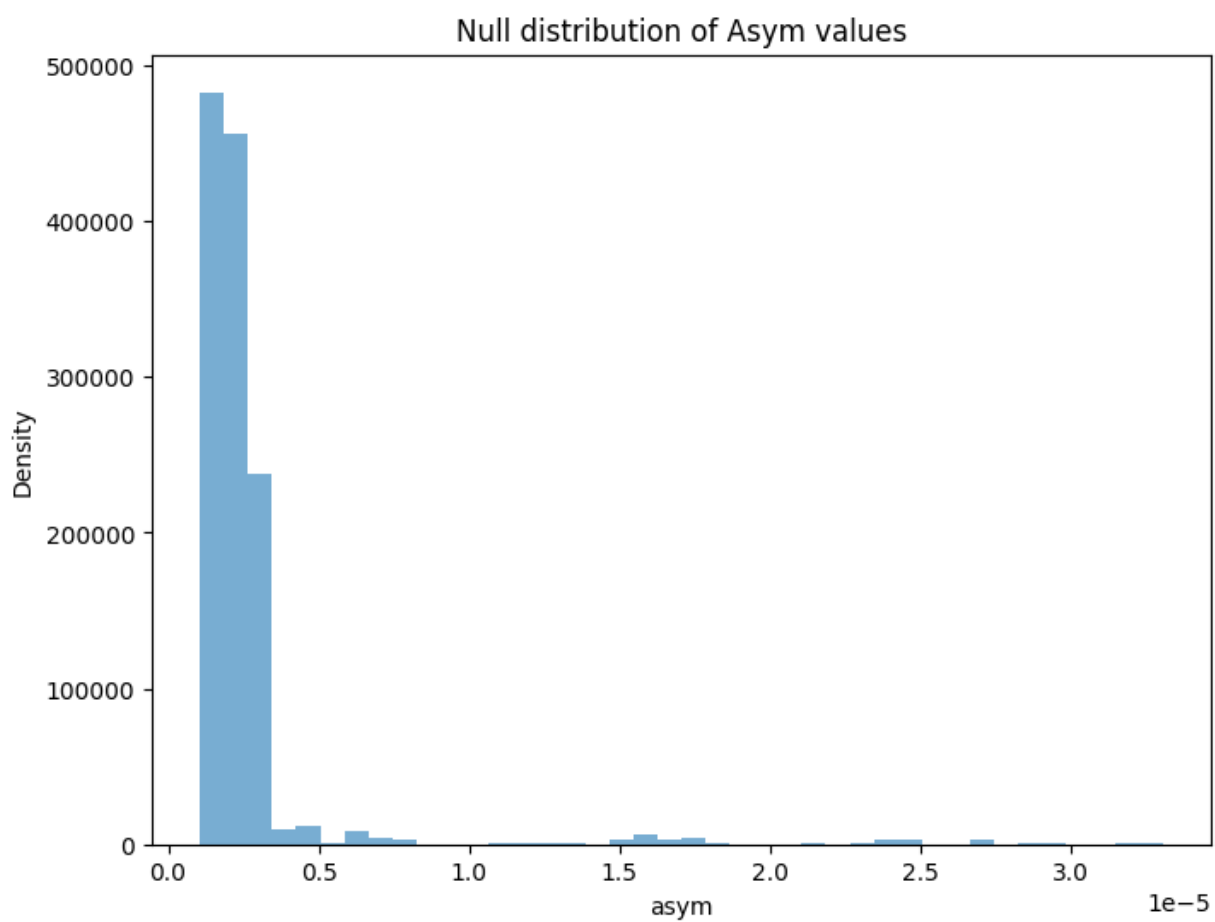
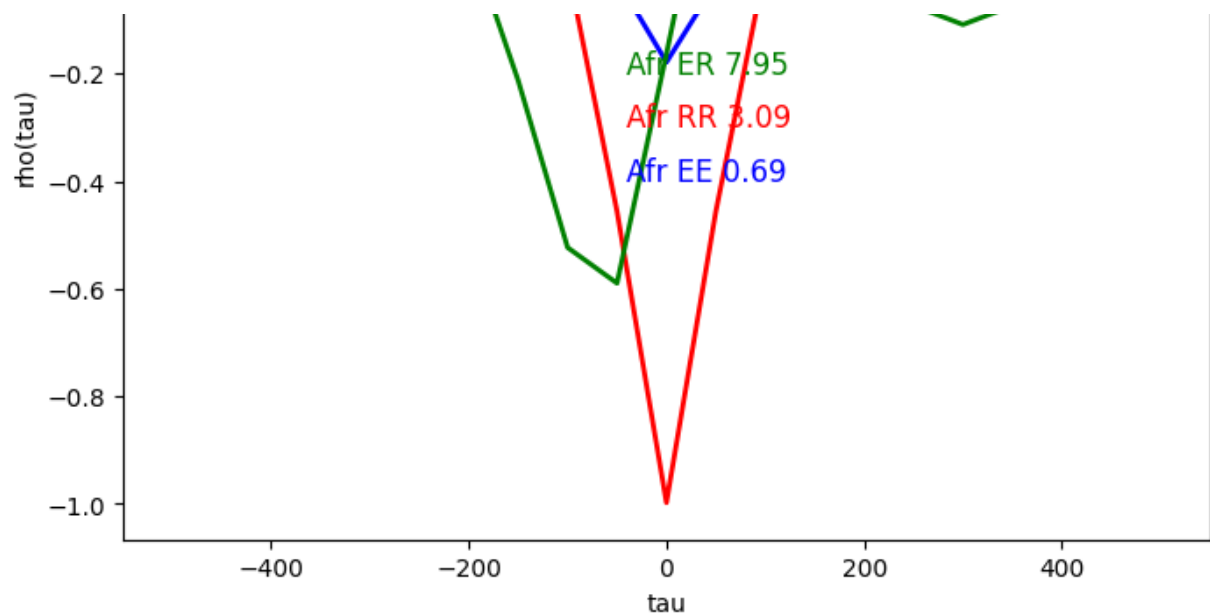
Double-click (or enter) to edit

Arrow-of-time asymmetry, dynamic systems example

Show code

```
Mounted at /content/drive
Variables in .mat file: dict_keys(['__header__', '__version__', '__global
0
100
200
300
400
500
600
700
800
900
```





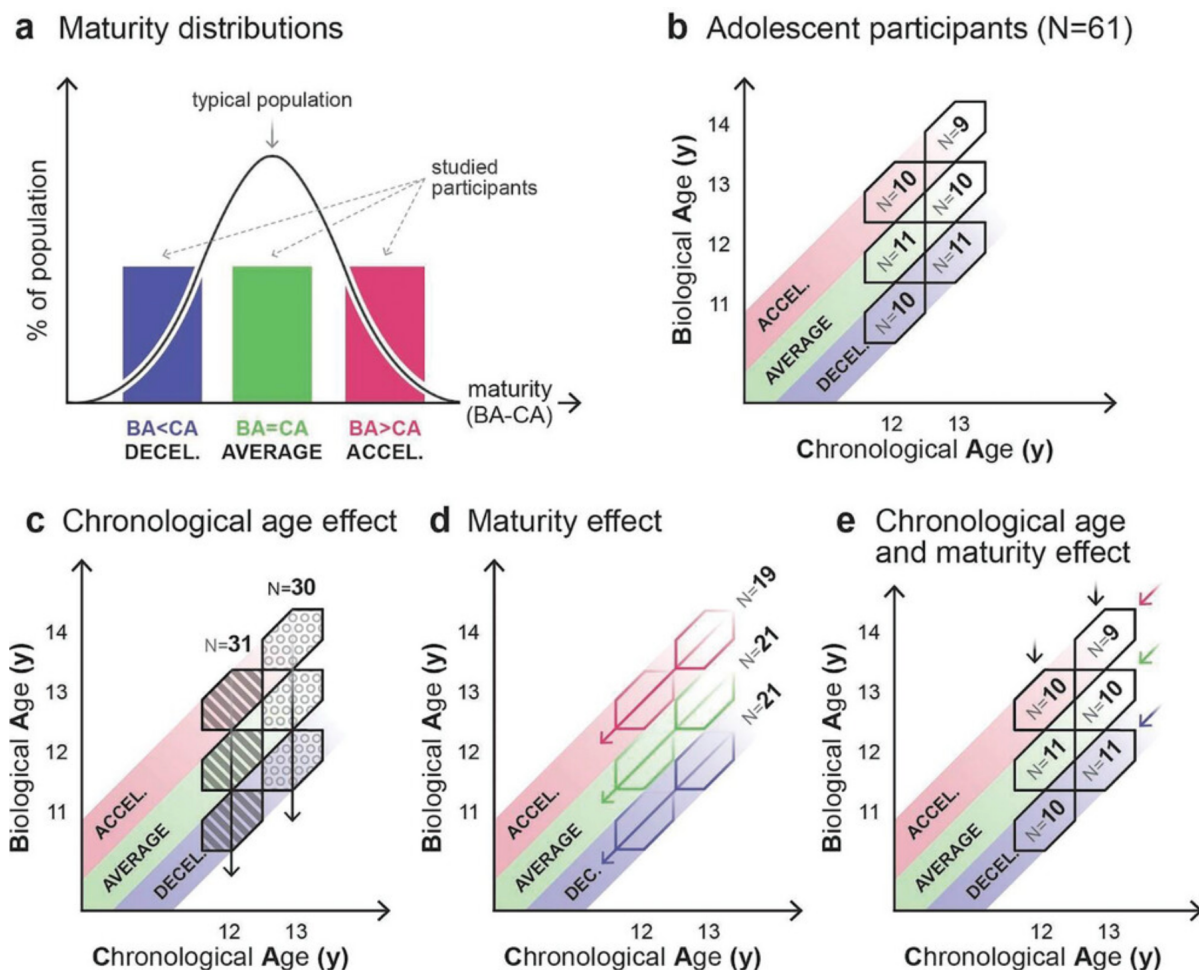
6. Null distributions for entropy

When computing measures of disorder or entropy, the shuffling approach faces some difficulties, because randomization typically inflates disorder or entropy.

To illustrate the problem, I use our recently published study on changes of entropy production in resting-state EEG signals measured from adolescent females.

The recordings were performed by our colleagues in Budapest and the analysis was developed and carried out in Magdeburg.

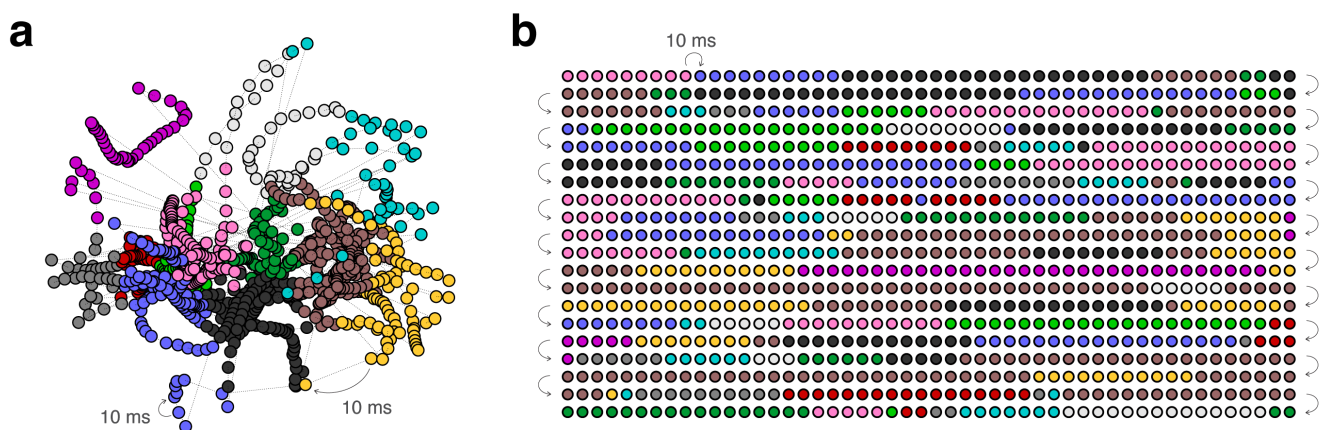
[Szakacs, Mutlu, et al. 2024](#)



Entropy production by trajectory of EEG macrostates

We recorded high-density EEG from resting subjects, established whole-scalp phase coherence patterns at 100 Hz, and analyzed the development of such patterns over time. Even though these coherence patterns develop in a temporally continuous manner, the dynamics of whole-scalp coherence can be approximated in terms of a sequence of distinct

macrostates. The level of detail in this description depends on the chosen number of categorical states.



Panel a shows a representative trajectory of 128-dimensional microstates, projected into 3 dimensions. Sequence duration is 10 s (or 1000 microstates). Note that the trajectory moves largely continuously (in 10 ms steps). The apparent jumps (straight dotted lines) are computational artefacts owed to the ambiguity of eigenvector signs.

Panel b illustrates the same sequence in terms of categorical macrostates distinguished by color.

Macrostate transitions

Having defined distinct "macrostates" (as categories or clusters of "microstates"), we established the statistics of macrostate 'transitions' for every recorded sequence of eigenvectors. Consider an observed sequence of N eigenvectors assigned to K distinct clusters. For simplicity, we made sequences cyclical by pretending that the first vector succeeded the last vector. From every such sequence, we established transition counts n_k^j , that is, the observed number of transitions from cluster of origin j to cluster of destination $k \neq j$.

$$n_k^j \quad \text{from } j \text{ to } k, \quad N_{trans} \equiv \sum_{k=1}^K \sum_{j \neq k} n_k^j$$

where N_{trans} is the total number of transitions. As any transition is *from* one state *to* another state, row and column sums in the matrices n_k^j are equal

$$\sum_k n_k^j = \sum_k n_j^k, \quad \forall j$$

Below is an example of an observed transition matrix. Note that this matrix is not exactly symmetric. This asymmetry could be accidental (due to the finite duration of the observed

sequence) or it could be systematic (due to non-equilibrium dynamics of brain activity).

We want to determine whether the observed degree of asymmetry is statistically significant!

To: **Transition matrix, asymmetric**

1	0	0	41	0	43	1	57	22	16	2	2	30
2	0	0	37	95	1	1	3	43	1	17	60	3
3	50	42	0	22	1	6	1	7	25	4	9	28
4	0	84	41	0	0	1	0	3	36	40	1	74
5	40	0	2	1	0	23	78	0	56	0	3	35
6	1	0	6	0	29	0	14	0	14	20	17	7
7	55	5	0	0	86	19	0	70	1	0	48	1
8	14	42	1	2	0	2	79	0	0	51	24	1
9	14	2	32	34	47	12	2	2	0	19	4	29
10	5	19	3	48	0	14	3	45	16	0	0	5
11	3	66	10	0	2	19	46	22	2	0	0	1
12	32	1	22	78	29	10	2	2	30	5	3	0

From: 1 2 3 4 5 6 7 8 9 10 11 12

✓ Entropy production

The commonly used expression for entropy production in non-equilibrium systems ([Schnakenberg, 1976](#)) is the weighted sum of imbalanced flows

$$\dot{S}_{tot} = \sum_{k \neq j} \left[w_k^j P_j - w_j^k P_k \right] \log \frac{w_k^j P_j}{w_j^k P_k}$$

where the sum is taken over all *unordered* pairs $k \neq j$.

Each unordered pair $k \neq j$ contributes

$$\dot{S}_{jk} = \dot{S}_{kj} = \frac{n_k^j - n_j^k}{N} \log \frac{n_k^j}{n_j^k}$$

to the total entropy production.

When no transitions were observed for a subset of ordered pairs $k \rightarrow j$, the associated transition flows vanish and the entropy production can be arbitrarily large. To avoid this issue

transition flows vanish and the entropy production can be arbitrarily large. To avoid this issue, we changed the value of all $n_j^k = 0$ to $n_j^k = 1$. This arbitrary choice lowered entropy production and set the numerical scale for its absolute value. Due to this arbitrary modification, only the relative values of entropy production are meaningful and interpretable.

Bootstrapping entropy-production, the wrong way

Show code

Nbootstrap 500
 Nsequence 20000

Observed Sequence length: 20000

Observed Entropy: 0.0254

[Bootstrap 100/500]

[Bootstrap 200/500]

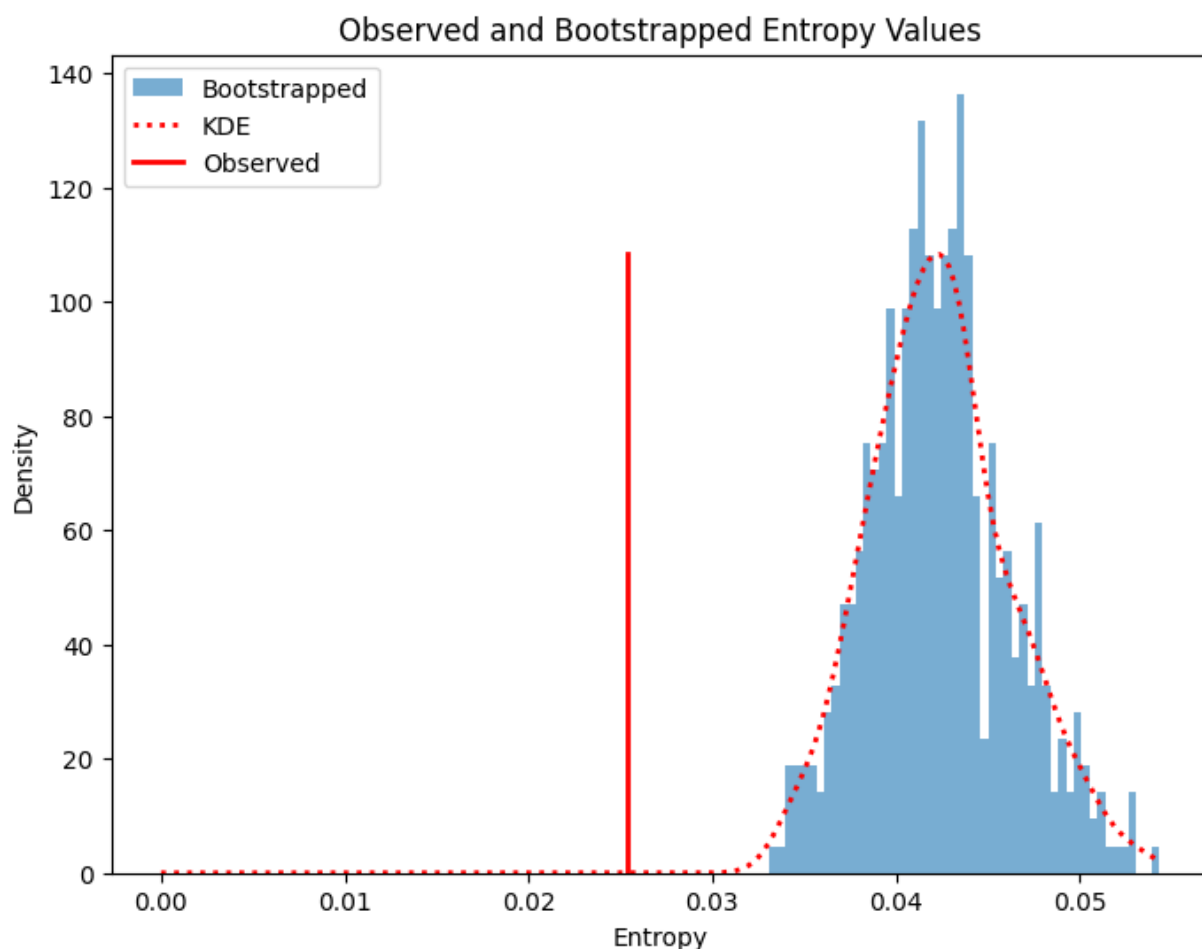
[Bootstrap 300/500]

[Bootstrap 400/500]

[Bootstrap 500/500]

Mean Entropy: 0.0424

Std Entropy: 0.0038



```
plot_observed_and_bootstrapped_entropy
def plot_observed_and_bootstrapped_entropy(Nbootstrap=500,
Nsequence=5000)
```

<no docstring>

✓ Bootstrapping entropy the right way

We find that bootstrapped entropy values are distributed **to the right** of our observed value. This is a surprise, as we would expect bootstrapped values to distribute on both sides (**left and right**) of the observed value.

The problem stems from the finite length of the observed trajectory. As we generate very long synthetic trajectories, the entropy value approaches the observed value, demonstrating that our bootstrapping procedure works.

Evidently, generating trajectories of finite length introduces noise, which contributes spurious entropy and inflates the bootstrapped values. How much noise does a trajectory of a certain length contribute?

We can answer this empirically, by generating trajectories from a symmetric matrix, for which the theoretical entropy production is exactly zero. Accordingly, any entropy production that we observe for a symmetric matrix must be due to noise.

The average amount of spuriously produced entropy can then be used to correct (left-shift) our bootstrapped distributions.

To: Transition matrix, asymmetric

1	0	0	41	0	43	1	57	22	16	2	2	30
2	0	0	37	95	1	1	3	43	1	17	60	3
3	50	42	0	22	1	6	1	7	25	4	9	28
4	0	84	41	0	0	1	0	3	36	40	1	74
5	40	0	2	1	0	23	78	0	56	0	3	35
6	1	0	6	0	29	0	14	0	14	20	17	7
7	55	5	0	0	86	19	0	70	1	0	48	1
8	14	42	1	2	0	2	79	0	0	51	24	1
9	14	2	32	34	47	12	2	2	0	19	4	29
10	5	19	3	48	0	14	3	45	16	0	0	5
11	3	66	10	0	2	19	46	22	2	0	0	1

..

12	32	1	22	78	29	10	2	2	30	5	3	0
----	----	---	----	----	----	----	---	---	----	---	---	---

From: 1 2 3 4 5 6 7 8 9 10 11 12

To: **Transition matrix, symmetric**

1	0	0	45.5	0	41.5	1	56	18	15	3.5	2.5	31
2	0	0	39.5	89.5	0.5	0.5	4	42.5	1.5	18	63	2
3	45.5	39.5	0	31.5	1.5	6	0.5	4	28.5	3.5	9.5	25
4	0	89.5	31.5	0	0.5	0.5	0	2.5	35	44	0.5	76
5	41.5	0.5	1.5	0.5	0	26	82	0	51.5	0	2.5	32
6	1	0.5	6	0.5	26	0	16.5	1	13	17	18	8.5
7	56	4	0.5	0	82	16.5	0	74.5	1.5	1.5	47	1.5
8	18	42.5	4	2.5	0	1	74.5	0	1	48	23	1.5
9	15	1.5	28.5	35	51.5	13	1.5	1	0	17.5	3	29.5
10	3.5	18	3.5	44	0	17	1.5	48	17.5	0	0	5
11	2.5	63	9.5	0.5	2.5	18	47	23	3	0	0	2
12	31	2	25	76	32	8.5	1.5	1.5	29.5	5	2	0

From: 1 2 3 4 5 6 7 8 9 10 11 12

In this way, we can establish the statistical significance of our observed entropy value.

Bootstrapping entropy-production, the right way

Show code

Nbootstrap

Observed Sequence length: 2537

Observed Entropy: 0.0254

[Bootstrap 100/500]

[Bootstrap 200/500]

[Bootstrap 300/500]

[Bootstrap 400/500]

[Bootstrap 500/500]

[Bootstrap 100/500]

[Bootstrap 200/500]

[Bootstrap 300/500]

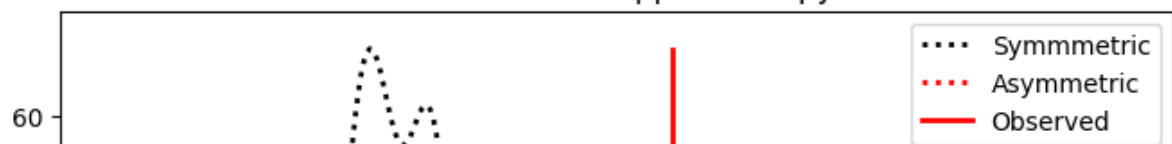
[Bootstrap 400/500]

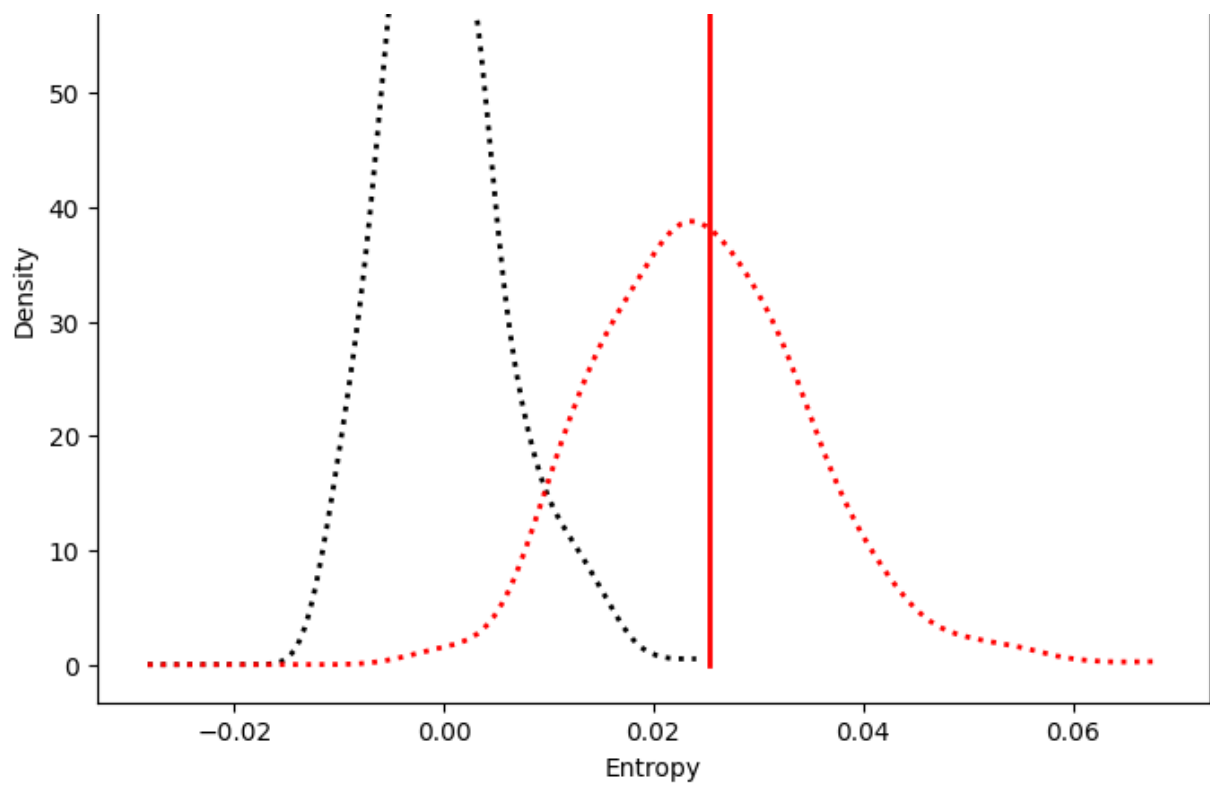
[Bootstrap 500/500]

Mean Entropy: 0.0530

Std Entropy: 0.0102

Observed and Bootstrapped Entropy Values





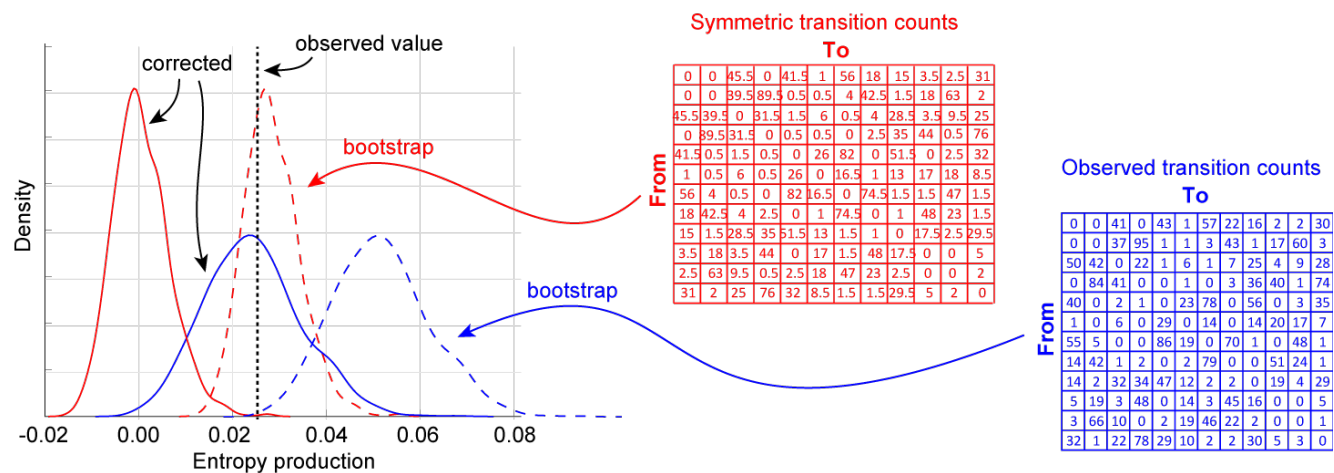
```
plot_observed_and_corrected_bootstrap
```

```
def plot_observed_and_corrected_bootstrap(Nbootstrap=500)
```

```
<no docstring>
```

Summary

This figure summarizes the corrections that are required for bootstrapping entropy production.



Summary

1. Bayes Factor
2. Bayesian Information Criterion (BIC)
3. False discovery corrections
4. Bootstrapping for standard error and confidence
5. Shuffling for null distributions

Following lectures

Lecture 7: Bootstrapping examples (Stepan Aleshin)

Lecture 8: GLM examples (Stepan Aleshin)

Lecture 9: Linear discriminant analysis (Jochen Braun)

Lecture 10: Multivariate LDA (Jochen Braun)